A Flexible Deviation from FIRE in the Sequence Space
[Conference draft: do not cite]

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February 15, 2026

Abstract

This paper proposes a flexible approach to departing from full information rational expectations (FIRE) in DSGE models using the sequence space framework. Building on Kohlhas & Walther (2021), we implement a reduced-form behavioral expectations process in which agents simultaneously overreact to current economic conditions and underreact to news. The approach nests several different expectations models as special cases while remaining agnostic on the precise source of belief frictions. We apply the framework to the two-asset HANK model of Auclert et al. (2020) and jointly estimate the model's dynamic and behavioral parameters on US business cycle data, using inflation expectations from the Survey of Professional Forecasters and the Michigan Survey to separately discipline household and firm expectations. Both sectors significantly depart from rational expectations, with firms updating quickly but extrapolating substantially from current conditions and households exhibiting extreme inattention. Extrapolation interacts synergistically with sticky expectations, tripling the peak investment response and quadrupling the peak inflation response to monetary policy shock, while helping generate the pattern of initial underreaction and delayed overreaction documented by Angeletos et al. (2021). Including extrapolation also improves the models forecasting properties relatives to the no-extrapolatio baseline.

Keywords: Behavioral expectations, HANK models, Sequence space, Monetary policy, Asymmetric attention, Business cycle estimation

JEL Classification: E31, E52, E70, D84

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1 Introduction

Can we depart from full information rational expectations (FIRE) in a flexible and standardized manner that is consistent with empirical evidence from macro, micro, and behavioral economics? This question has gained renewed importance as the recent heterogeneous agent New Keynesian (HANK) literature emphasises the need to match both microeconomic evidence and macroeconomic dynamics, particularly hump-shaped impulse responses to monetary policy shocks in empirical data, simultaneously. While Auclert et al. (2020) propose sticky expectations as a solution to generate persistence in macro humps, this approach tends to reduce amplification—a potentially problematic feature highlighted by Albuquerque et al. (2025). Futher, the approach also ignores microeconomic evidence on deviations from rational expectations that agents not only update their information and expectations infrequently, but also in forming their expectations they extrapolate from current conditions.

Taking stock, this paper proposes a semi-structural departure from rational expectations based on the micro-founded empirical framework of Kohlhas & Walther (2021). Their reduced-form specification captures how actual expectations deviate from rational expectations through two channels: extrapolation from current conditions and underreaction to news. Our first contribution is implementing this behavioral expectations process in the sequence space framework for the solution and estimation of DSGE and HANK models developed by Auclert et al. (2021), allowing for tractable incorporation of non-rational expectations into complex, quantitatively rich models. As we show when laying out our theoretical framework, our implementation of the process building on Kohlhas & Walther (2021) is best understood as a generalisation of sticky-expectations and informational rigidity models to a richer case such that whenever agents infrequently update their information set, they incur a further bias from extrapolating current macroeconomic conditions.

Our approach to expectation formation offers several advantages. First, it provides a clear mapping from the empirical applied literature evidencing simultaneous under-reactive and over-reactive components to agents expectations to a generalised implementation in state-of-the-art macro models, including but not limited to HANK models. Second, it allows agents to simultaneously overreact to outcomes and underreact to news,[1] capturing important behavioral patterns documented in survey data. Third, it nests as restrictions key different expectation models as special cases of a generalised behavioural expectations process, enabling direct model comparisons relative to matching empirical IRFs and business cycle evidence, theoretically regimenting the *wilderness* in models of expectations formation in macroeconomics in a way that is at the same time parsimonious and integrable with macro and micro-data, for meaningful comparison of models in terms of empirical applicability and theoretical implications for the modelling of macroeconomic transmission. These include FIRE beliefs, asymmetric attention, sticky or noisy information when the *news revision* parameter is positive and the *extrapolation* one is zero, and extrapolative expectations when the opposite is true. Fourth, it represents near-rational or internally-rational expectations as in Adam & Marcet (2011), where agents behave optimally conditional on their subjective beliefs, expectations converge[2] to rational expectations in the long run, and the use of subjectively available information in forecasting future aggregates is rational, in the sense that a law of iterated expectations hold and future forecast errors are orthogonal to agents' projections.

We demonstrate our approach by applying it to the two asset HANK model of Auclert et al. (2020). The authors in that paper advocate for sticky expectations in order to to produce empricially consistent hump shaped impulse responses of macroeconomic aggregates while maintaining empirically realistic marginal propensities to consume. In so doing their IRF matching approach calls for a considerable degree of stickiness of expectations for households while maintaining rational expectations outside of the household sector. We allow for departures from rational expectations outside of the household sector and jointly estimate the models dynamic and behavioral parameters on US business cycle data using a full information Bayesian maximum likelihood approach. By including inflation expectations data from the Survey of Professional Forecasters and Michigan Survey we are able to discipline our behavioral parameter estimates. Our approach indicates that both households and non-household

[1] Understood as shocks or changes in the information sets on which agents project future aggregates between consecutive time periods

[2] Under the most reasonable parameterizations and assumptions.

(firms) sectors significantly depart from the rational expectations benchmark. Both simultaneously extrapolate to outcomes today and under react to news about tomorrow consistent with the evidence of Kohlhas & Walther (2021). Non-household sectors react to news relatively quickly in line with the extant empircal evidence (e.g. Coibion & Gorodnichenko (2015)). However we estimate households to be close to unreactive to news about the future in the model; inconsistent with the micro-evidence. As in Albuquerque et al. (2025), the inclusion of the extrapolative terms creates greater amplification in the hump shaped response to IRFs. We also re-estimate the parameters with extrapolation turned off and compare the properties of the model to the one with extrapolation. Overall we find that including extrapolation reduces the estimates of stickiness on the firm side of the model, lowers investment adjustment cost estimates and produces better forecasting accuracy.

1.1 Related Literature

This paper relates to literatures spanning behavioral macroeconomics, particularly the role of expectations and informational frictions, heterogeneous agent business cycle models, and expectation formation. A growing body of empirical evidence documents systematic deviations from rational expectations in survey data and provides suggested micro foundations for those deviations. Coibion & Gorodnichenko (2015) find evidence of information rigidity in the expectations formation process akin to the sticky expectations of Mankiw & Reis (2003), showing that forecast errors are predictable using past information. Bordalo et al. (2018) propose diagnostic expectations where agents overweight recent news. Adam et al. (2024) studies the household response to house price movements and documents initial underreaction followed by overreaction and explains this through extrapolative expectations. Importantly, Angeletos et al. (2021) document a new stylised fact that unifies much of this seemingly conflicting evidence: in response to the main shocks driving the business cycle, expectations initially underreact but subsequently overshoot actual outcomes. They show that this pattern of dynamic overshooting speaks in favour of models combining dispersed, noisy information with overextrapolation.

Departures from rational expectations have also been the focus of research into financial markets and asset pricing. Bordalo et al. (2020) demonstrate that diagnostic expectations can simultaneously explain momentum and reversal patterns in asset prices and systematic forecast errors in macroeconomic surveys, finding that individual forecasters typically overreact to news while consensus forecasts underreact relative to full-information rational expectations. We build upon the specific reduced-form approach of Kohlhas & Walther (2021), who document that households, firms, and professional forecasters simultaneously underreact to news and overreact to current conditions, embedding many of the features or outcomes of the above expectations processes.

Our approach is implemented in the sequence space using the methods of Auclert et al. (2021) and further generalised by Bardóczy & Guerreiro (2024) to allow for more arbitrary expectations processes. This approach embeds and assumes the concept of internal rationality as put forward by Adam & Marcet (2011). Bardóczy & Guerreiro (2024) combine noisy information, diagnostic expectations, and long-term memory to study the importance of the expectations channel for the design of unemployment insurance programs. We provide a specific simple and flexible application of their method and approach. Angeletos et al. (2025) study the transition from RANK to HANK models without the assumption of full information rational expectations. Their work highlights how the interaction between household heterogeneity and departures from FIRE shapes the transmission of aggregate shocks, emphasising that the quantitative implications of HANK models depend critically on the assumed expectations process. Our paper complements their analysis by providing an estimated quantitative framework that disciplines the expectations process with survey data, and by showing how the specific combination of sticky expectations and extrapolation affects both the amplification and persistence of impulse responses in HANK models.

Relating more broadly to the HANK literature, Auclert et al. (2020) argue that departures from rational expectations are key to simultaneously matching the macro and micro evidence in the calibration of HANK models. They rely upon sticky expectations in their application, which we compare to both quantitatively in terms of model estimation against both empirical IRFs and US business cycle data and theoretically, as the proposed behavioural expectations model robustly nests and extends

their focus on macro-inattention by incorporating extrapolation from current conditions as a source of further bias, above and beyond the Calvo-rigidity, whenever agents update their information sets.

Finally, this paper contributes to a growing literature that uses survey expectations data as observables in the Bayesian estimation of DSGE models. Pioneering work by Del Negro & Eusepi (2011) showed that including SPF inflation expectations among the observables provides strong discriminating power between alternative models of expectations formation, significantly improving model fit and helping to pin down the behaviour of latent expectation processes. Del Negro & Schorfheide (2013) demonstrated that incorporating survey forecasts into DSGE model estimation improves out-of-sample forecasting performance. Slobodyan & Wouters (2012) estimate a medium-scale DSGE model with adaptive learning and find that the beliefs implied by learning are more closely related to survey evidence on inflation expectations than those under rational expectations, while also reducing the estimated persistence of exogenous shock processes. Our contribution to this literature is twofold. First, we estimate a HANK model — rather than the representative agent models that dominate the existing literature on estimation with survey observables — using both professional forecaster and household inflation expectations to discipline the behavioural parameters. Second, by including expectations from both the SPF and the Michigan Survey, we are able to separately identify the degree of information rigidity and extrapolation on the firm and household sides of the model.

2 Generalised Expectations Framework

Our proposed framework captures the key insight from Kohlhas & Walther (2021) that agents may simultaneously overreact to current economic conditions while under reacting to news about future economic conditions. We aim to do so in a simple and flexible way using a minimal amount of reduced form parameters.

To do so we start from the reduced form regression equations of Kohlhas & Walther (2021) that expresses the forecast errors of agents about an input x, k period ahead, as a function of observations of x today and news about future x, which in their paper[3], is proxied empirically by the average forecast $\bar{f}$ of x in the current and previous period x.

$$x_{t+k} - f_{it}x_{t+k} = \alpha_i + \gamma x_t + \delta(\bar{f}_t x_{t+k} - \bar{f}_{t-1}x_{t+k}) + \varepsilon_{i,t|t+k} \tag{1}$$

The parameter γ captures extrapolation from current conditions (the "Extrapolation" effect), while δ captures the reaction to news (the "News" effect). Statistically significant differences from zero of either γ or δ is a rejection of the rational expectations hypothesis. Studies e.g. Coibion & Gorodnichenko (2015), consistently find evidence that δ is statistically significant from and greater than zero even for professional forecasters. This indicates a tendency to underreact to news about future economic conditions. Kohlhas & Walther (2021) further demonstrate that not only is $\delta > 0$ consistently across surveys and agents types but simultaneously γ is consistently and statistically significantly less than zero. This indicates a tendency of agents to simultaneously overreact to x_t today while underreacting to new about future x_{t+k} . Coibion & Gorodnichenko (2015) show how for the case $\gamma = 0$ the reduced form of equation 1 is consistent with either sticky expectations where agents update their forecasts infrequently or with a noisy information setup where agents see x imperfectly i.e. with measurement error. For the more general case $\gamma \neq 0$ Kohlhas & Walther (2021) show how the reduced form of equation 1 is consistent with asymmetric attention whereby agents pay different levels of attention to different components of news about x. Our approach is silent on the particular micro founded deviation from rational expectations. However, in so doing our reduced-form approach is sufficiently flexible to capture several prominent behavioral models that have emerged in the literature. Beyond the sticky information and noisy information frameworks explicitly discussed by Coibion & Gorodnichenko (2015), our specification can accommodate the essence of diagnostic expectations as in Bordalo et al. (2018), where agents overweight recent news relative to its true likelihood. In the diagnostic expectations framework, agents form beliefs by overweighting the probability of states that have become more likely in light of recent data, which can generate both over-reaction to salient

[3]As is typical in this literature

current information and systematic forecast errors that depend on the nature of recent news. The γ reflects the over-weighting of current salient information. The framework is also consistent with models of extrapolative expectations, where agents project recent trends into the future. Where γ and δ are allowed to vary by x, it may also be used to approximate rational inattention models where agents optimally choose to remain uninformed about certain aspects of the economic environment due to information processing costs. This flexibility allows our empirical specification to remain agnostic about the precise cognitive or informational friction driving departures from rational expectations, while still capturing the key asymmetric patterns in attention and updating that characterize different expectations processes and deliver more realistic macroeconomic dynamics.

2.1 Implementation in the Sequence Space and the Behavioural DSGE Solvent

We will now go through how to map the reduced form equation 1 into the sequence space such that we can apply the framework of Auclert et al. (2021) under this more generalised expectations case. For ease of exposition we will focus on the case of a constant δ and γ which can be relaxed as we show in the appendix. In the appendix we also show how to map deviations about expectations in growth rates into level or log level differences from steady state. These parameters can also vary by input x as required or supported by the empirical evidence.

2.1.1 Behavioural Forecast and Forecast Mappings

The first step in mapping equation 1 into the sequence space is to derive a forecasting equation for x. First we consider taking expectations[4] over equation 1 assuming no correlation of $\epsilon_{i,t|t+k}$ across agents. Doing so and rearranging results in the following expression for the average k period ahead forecast of x:

$$\bar{f}_t x_{t+k} = c + \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1} x_{t+k} + E_t^*[x_{t+k}] - \gamma x_t\right) \tag{2}$$

where we note $\bar{f}_t x_{t+k} = E[f_{it} x_{t+k}]$ and E^* is the mathematical expectations operator conditional on the full information set for for x. Since in the sequence-space this information set coincides with the time zero one (and with the one at all subsequent times, i.e. a stationary filtration), this is equivalent to perfect foresight or actual outcome x, and we thus suppress the operator in the following derivations. Over and above certainty equivalence we assume: (1) agents have unbiased expectations in the steady state i.e. $c = \frac{\gamma}{1+\delta} x_{ss}$; (2) that for some large T agents expect a return[5] to the steady state x_{ss} and (3) that for agents the law of iterated expectations holds, such that agents do not expect to make forecast errors. These assumptions combine to ensure agents act rationality given their internal beliefs and are internally rational as described by Adam & Marcet (2011), and are expounded in (Appendix 5.2). Under these technical assumptions it can be shown (Appendix 5.1) that at a 1st order perturbation expansion around the steady state, the LoM for the *behavioural* forecast follows:

$$\bar{f}_t dx_{t+k} = \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1} dx_{t+k} + dx_{t+k} - \gamma dx_t\right).$$

This can be conveniently rewritten as:

$$\bar{f}_t dx_{t+k} = \overbrace{\frac{\delta}{1+\delta}}^{\text{Stickyness } P(\Omega_t = \Omega_{t-1})} \bar{f}_{t-1} dx_{t+k} + \overbrace{\frac{1}{1+\delta}}^{\text{Pr. update}} (dx_{t+k} - \gamma dx_t).$$

The last equation lays the ground for the intuition on how the behavioural model flexibly nests and enriches the sticky or inattentive expectations case in a way such to make it possible to simultaneously

[4] A natural application would be to take expectations over all households. As long as expectations are permanent one could also consider taking expectations for different groups of households. One of the advantages of the sequence space Jacobian approach laid out by Auclert et al. (2021) is in working with aggregated outcomes at the block level. Combining policy function choices with distributional dynamics. Averaging in this way is consistent with this approach.

[5] This implies that equation 2 holds for $k < T$ and that the final model must converge to x_{ss} in general equilibrium.

account for the macro-humps (or delayed response) characterising monetary policy transmission and for stronger endogenous amplification of external shocks. This is explored in greater detail in the next section (a recursive forecast error view); the main idea however is that the forecast formed at time t can be viewed as a moving average of the forecast formed in the previous period – resulting from Calvo-style rigidities or stickyness ($\delta/(1+\delta)$ on information updates – and the innovation against such forecast for that fraction of agents who update at the beginning of this period ($1/(1+\delta)$). Contrary to the pure sticky case, those who update still make a forecast error through extrapolation $-\gamma dx_t$ from the economic conditions at the time they update their beliefs.

We can express the above, equivalently, in terms of a law of motion for the HH forecast error on the aggregate inputs at some fixed period $t+k$, for each such period. The forecast error is naturally defined as:

$$h_t^{t+k} = dx_{t+k} - \bar{f}_t dx_{t+k}$$

Using the law of motion for the forecast, This implies a law of motion for the forecast error, for all t, k:

$$h_t^{t+k} = \left(\frac{\delta}{1+\delta}\right) dx_{t+k} + \frac{1}{1+\delta}(\gamma dx_t - \delta \bar{f}_{t-1} dx_{t+k})$$

$$h_t^{t+k} = \left(\frac{\delta}{1+\delta}\right) dx_{t+k} - \frac{\delta}{1+\delta} dx_{t+k} + \frac{1}{1+\delta}(\gamma dx_t + \delta h_{t-1}^{t+k})$$

$$\underbrace{h_t^{t+k}}_{\text{posterior forecast error}} = \frac{\delta}{1+\delta} \underbrace{h_{t-1}^{t+k}}_{\text{prior forecast error}} + \frac{1}{1+\delta} \underbrace{\gamma dx_t}_{\text{innovation under extrapolative expectations}}$$

The last equation tells us that under the proposed behavioural model, the forecast error h_t^{t+k} at time t on some arbitrarily timed aggregate input x_{t+k}, for any k, is a linear combination or moving average between the forecast error in the previous period and extrapolation from current conditions (γdx_t), where the weighting is $\delta/(1+\delta)$. The evolution of the forecast error over time can be then viewed (ex-post, or from the econometrician's point of view) as the output of a filtering algorithm, where agents update their prior forecast error given information gathered from extrapolation. Whether the extrapolation from current conditions or the last period forecast error dominates today's forecast error depends on the degree of stickiness or, in our framework, under reactions to news revision: for some bounded value γ, as stickiness (inattention)in the limit $\delta \to \infty$, the forecast error law of motion is approximated by a unit root process, and extrapolation from current conditions ceases to matter. This provides an intuition on the equivalence to Auclert et al. (2020): infinitely sticky expectations correspond to a case in which agents fail to update their information sets *almost surely*. More formally, modelling the updating process for inattentive agents as a sequence of i.i.d. Bernoulli trials with success probability $(1-\theta)$, where

$$\delta = (1-\theta)\sum_{s=0}^{\infty} \theta^s s = \frac{\theta}{1-\theta}$$

is the expected number of periods elapsed before agents update their information sets and expectations,[6] it follows the law of motion for the forecast error can be equivalently written as:

$$h_t^{t+k} = \theta h_{t-1}^{t+k} + (1-\theta)\gamma dx_t$$

Conversely, when agents pay perfect attention and expectations are purely extrapolative, the forecast error displays no inertia or sluggishness and is a measurable function – a measurement or jumper in DSGE vernacular – of the underlying current aggregate state only. The main implication of this is that the two behavioural channels interact in complementary ways – i.e. their is a *synergic* interaction

[6] The number of periods s elapsed for a single household before it updates its information set and, with the assumed law of large numbers, the number of households in the cross-section that use expectations formed $s = 0, ..$ periods ago (i.e. have failed to update since) are modelled by the geometric distribution with *failure* probability θ,

– in shaping agents forecasts and ultimately the propagation of aggregate shocks to aggregate fluctuations around the steady state. In particular, the news-revision parameter is key for sustaining inertia or sluggishness in forecast revisions over time, while extrapolation (under $\gamma < 0$ or over-reaction to current conditions) makes the forecast error depend pro-cyclically on current conditions. The novelty of the framework in accommodating these two separate, yet interacting channels is key for the ability of the model to reconcile stronger endogenous amplification in the face of identically distributed shocks at similar levels of inattention or informational rigidity, with the macro-humps familiar from simulation and estimation with inattentive agents. Finally, returning to internal rationality, for a *bounded* positive δ (implying θ within the unit circle), along a non-stochastic path where aggregate shocks have settled to zero, the forecast error converges to zero for a large enough t (see assumption 4 in appendix 6.2).

Backward-solution and behavioural forecast mapping. To back out the general equilibrium SSJ solvent, exploiting a procedure equivalent to that in Bardóczy & Guerreiro (2024), we begin by using the recursive law of motion for the behavioural forecast, the economic intuition for which we laid out above, to construct a mapping $\mathcal{E}_\tau : d\mathbf{X} \to X^{e,\tau} = \Lambda_\tau d\mathbf{X}$ from the space of true aggregate inputs sequences $d\mathbf{X}$ to the space sequences of of timeτ expectations on the path of inputs $d\mathbf{X}^{e,\tau}$. Starting at the steady state we now consider how the forecast of $x_k = x_{ss} + dx_k$ evolves as we move through time from period 0 to period k. The evolution is as follows:[7]

$$\bar{f}_0 x_k = x_{ss} + \frac{1}{1+\delta} dx_k - \frac{\gamma}{1+\delta} dx_0$$

$$\bar{f}_1 x_k = x_{ss} + \left(\frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} \right) dx_k - \frac{\gamma}{1+\delta} dx_1 - \frac{\gamma\delta}{(1+\delta)^2} dx_0$$

$$\bar{f}_t x_k = x_{ss} + \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{t-j} \frac{1}{1+\delta} dx_k - \gamma \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{t-j} \frac{1}{1+\delta} dx_j \tag{3}$$

where for the period $t < k$ the first sum represents the news effect which grows over time and asymptotically converges to perfect foresight for $\delta > 0$.[8] With bounded δ or inattention $\theta = \delta/(1+\delta)$ so that however small there is always a nonzero probability agents update their information sets, almost-sure asymptotic rationality is equivalent to the idea that the set of households who always fail to update is a zero-measure set.[9] The second sum captures the extrapolation effect, where agents extrapolate from the sequence of realisations of x observed up until period k with greatest weight given to more recent observations. We see that forecasts for x are therefore neatly linearly decomposable as a function of the histories of x and news about x in the future.

[7] This is shown in the Appendix 7.1

[8] This follows because, noting that *for any t:*

$$\frac{1}{1+\delta} \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{t-j} = \frac{1}{1+\delta} \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{j},$$

then it follows from the Neumann Series Lemma (NSL) that

$$\lim_{t\to\infty} \frac{1}{1+\delta} \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{t-j} = \frac{1}{1+\delta} \sum_{j=0}^{\infty} \left(\frac{\delta}{1+\delta} \right)^{j} = \frac{1}{1+\delta} \left(1 - \frac{\delta}{1+\delta} \right)^{-1} = 1$$

[9] Effectively, almost-sure asymtptotic rationality follows from a direct application of the first *Borel-Cantelli* Lemma, under our result in the previous footnote:

$$\sum_{j}^{\infty} P(A_j) < \infty \to P(\liminf_j A_j) \leq P(\limsup_j A_j) = 0,$$

when letting, in an appropriately defined measured space, A_j the measurable set of households who sequentially fail to update for j consecutive periods – so that in the limit the appropriate concept for never-updating households is the limit inferior of the sequence of sets – and P the probability measure on it.

This backward-solution for the linearised forecast in terms of observable history and news forms the basis for the mapping from actual aggregate input sample paths to time-varying expectations thereon we employ to construct the sequence-space DSGE solvent. Before making this and the associated mapping Λ_k explicit, note the above can be rewritten as (letting again $d\bar{f}_t x_k = \bar{f}_t x_k - x_{ss}$):

$$d\bar{f}_t x_k = \frac{1}{1+\delta} \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta} \right)^{t-j} (dx_k - \gamma dx_j)$$

$$d\bar{f}_t x_k = (1-\theta) \sum_{j=0}^{t} \theta^{t-j} (dx_k - \gamma dx_j)$$

Where we have used our observation on how the news-revision coefficient δ identifies the macro-inattention or stickiness parameter from above, along the lines of Coibion & Gorodnichenko (2015). Similarly to the setup in Auclert et al. (2020), we can thus equivalently view the average behavioural forecast governed by the outlined laws of motion as the average of forecasts in a cross-section of permanently heterogeneous agent types differing in the time they last updated their expectations, where θ^{t-j} is the probability of having failed to update for the j-periods preceding time t when t comes, and under a LLN the measure of the set of agents or fraction of agents who last updated their expectations at $t-j$. Relative to the sticky expectations, cementing the point made in the context of the recursive law of motion, as agents update their information sets and forecasts in the behavioural framework they incur a further bias stemming from pro-cyclical extrapolation from current conditions at the time when they happen to (last) update their beliefs $-\gamma x_k$. In other words, while news are sluggishly but progressively incorporated by agents just like in sticky expectations from informational rigidities, the path to asymptotic perfect foresight is disturbed at any point in time by the state of economic aggregates at the time informational update happens. Suppressing such extra bias, as already noted in Coibion & Gorodnichenko (2015), yields the nested purely sticky case.

2.1.2 Generalisation of the Sequence-Space Jacobian to linear forecast mappings

Bardóczy & Guerreiro (2024) show how generalised expectations processes can be mapped into the sequence space building on the objects derived in Auclert et al. (2021). In the approach of Auclert et al. (2021) the economic model is broken down into a series of decision problems that map block level inputs X to outputs $Y(X)$ in the sequence space. For example household aggregate household consumption C is determined by the real interest rate r, real wage w and distribution of households D in the canonical Aiyagari-Bewley and Krussel-Smith models. Under a first-order approximation around the non-stochastic steady state and certainty equivalence, deviations in dY can be expressed as a product of dX and a Jacobian $J \in R^{T \text{x} T}$ that linearly maps dX to dY. By breaking out Y as a function of realisations of X and forecasts of X^e, $Y(X, X^e)$, Bardóczy & Guerreiro (2024) show how under the assumptions made we can derive a generalised Jacobians $\hat{J}$ as a function of the rational expectation Jacobian J derived in Auclert et al. (2021) and a sequence of forecast for $X^e \in R^{T \text{x} T}$:

$$dY = J dX^{e,0} + \sum_h R_h (dX^{e,h} - dX^{e,h-1}) \tag{4}$$

$$\text{with} \quad R_h = \begin{pmatrix} 0 & 0_h^{'} \\ 0_h & J \end{pmatrix}$$

where R_h shifts the rational expectations Jacobian diagonally by the horizon h. Equation 4 intuitively shows outcomes of Y through the sequence space as depending upon a series of unanticipated shocks to outcomes and expectations of X. For example at time zero a shock hits the economy which might change X today and causes agents to update their expectations of future X^e. The effect of this can be computed with the first row of the rational expectations Jacbobian J. If no further shocks happen and there are no revisions to expectations then the we simply continue to apply the rational expectations Jacobian as we move through time (down the rows of J). When further shocks happen or expectations are revised and this is unanticipated (as is the case under the assumption, consistent

with internal rationality, that agents do not forecast their future forecast errors or revision) it is equivalent to another set of unanticipated shocks to which we again can apply the first row of the rational expectations Jacobian J and again move down the rows as we move through time.

Our assumptions and equation 3 tells us forecasts of X only depend upon the the actual realisations X. When this is the case we can define a generalised behavioural expectations Jacobian $\hat{J}$ as a function of those forecasts and the rational expectations Jacobian J. This $\hat{J}$ is independent of any specific path X and conveniently allows us to compute $\hat{J}$ at the block level in partial equilibrium. To do so we we must define forecast matrices $\Lambda_k \in R^{TxT}$ that map the realized (perfect foresight) price path deviations dX into a sequence of forecasts $dX^{e,k}$. With those in hand we can define our behavioural expectations Jacobian $\hat{J}$

$$\hat{J} = J\Lambda_0 + \sum_h R_h(\Lambda_h - \Lambda_{h-1}) \tag{5}$$

where $dX^{e,0} = \Lambda_0 dX$ and $dY = \hat{J}dX$.

2.2 Generalised SSJ solvent with behavioural forecast mapping

The construction of an appropriate linear mapping Λ_h follows directly from the backward solution to the recursive law of motion for the behavioural forecast given in equation 3. Each Λ_k can be subdivided into 4 distict regions. The top left region is the identity matrix I_{h+1} representing the fact agents see inputs today and in history perfectly. Below the identify matrix is A_h which is the extrapolative effect that describes the effect of each observed x (moving along the columns) on forecasts of future X (moving down the rows). The bottom right region is the news effect which is a diagonal matrix B_h.

$$\Lambda_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ \frac{-\gamma}{1+\delta} & \frac{1}{1+\delta} & 0 & 0 & \cdots & 0 \\ \frac{-\gamma}{1+\delta} & 0 & \frac{1}{1+\delta} & 0 & \cdots & 0 \\ \frac{-\gamma}{1+\delta} & 0 & 0 & \frac{1}{1+\delta} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{-\gamma}{1+\delta} & 0 & 0 & 0 & \cdots & \frac{1}{1+\delta} \end{pmatrix}$$

$$\Lambda_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ \frac{-\gamma\delta}{(1+\delta)^2} & \frac{-\gamma}{1+\delta} & \frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} & 0 & \cdots & 0 \\ \frac{-\gamma\delta}{(1+\delta)^2} & \frac{-\gamma}{1+\delta} & 0 & \frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{-\gamma\delta}{(1+\delta)^2} & \frac{-\gamma}{1+\delta} & 0 & 0 & \cdots & \frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} \end{pmatrix}$$

$$\Lambda_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \frac{-\gamma\delta^2}{(1+\delta)^3} & \frac{-\gamma\delta}{(1+\delta)^2} & \frac{-\gamma}{1+\delta} & \frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} + \frac{\delta^2}{(1+\delta)^3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{-\gamma\delta^2}{(1+\delta)^3} & \frac{-\gamma\delta}{(1+\delta)^2} & \frac{-\gamma}{1+\delta} & 0 & \cdots & \frac{1}{1+\delta} + \frac{\delta}{(1+\delta)^2} + \frac{\delta^2}{(1+\delta)^3} \end{pmatrix}$$

For arbitrary h, the pattern generalises to:

$$\Lambda_h = \begin{pmatrix} \mathbf{I}_{h+1} & \mathbf{0}_{(h+1)\times(T-(h+1))} \\ \mathbf{A_h} & \mathbf{B_h} \end{pmatrix} \tag{6}$$

where the block-matrices $\mathbf{A_h}, \mathbf{B_h}$ for any h are generated by the function:

$$\mathbf{A}_{h(i,j)} = \begin{cases} \frac{-\gamma\delta^{h-j}}{(1+\delta)^{h-j+1}} & \text{for } j = 0, 1, \ldots, h \\ 0 & \text{otherwise} \end{cases}$$

,

$$\mathbf{B}_{h(i,j)} = \begin{cases} \sum_{l=0}^{h} \frac{\delta^l}{(1+\delta)^{l+1}} & i = j \\ 0, & i \neq j \end{cases}$$

By construction, for any time h at which the average forecast is formed, i.e. induced by the mapping, the (generally rectangular) block-matrix $\mathbf{A_h}$ collects the coefficients on the observable history from time zero through to h, corresponding to the pass-through $-\gamma$ to the forecast from current conditions at each time $t = 0, ..., h$ at which a subset of agents has (last) updated their information sets and expectations, weighted by the the measure $(\delta/(1+\delta))^{h-j}$ of such set of agents, i.e. the probability that they have not updated since. By contrast, block-matrix $\mathbf{B_h}$ collects along the main diagonal the news-revision coefficients on the each forecast input at time h, corresponding to the combined probability of agents learning about news on the forecast input at times $t = 0, ..., h$, or equivalently to the IRF of the forecast on some arbitrary input, viewed as a function of time, to a time-zero impulse to such input.[10]

2.3 General Equilibrium Solvent

The preceding sections have derived the behavioural forecast mappings Λ_h and the resulting behavioural expectations Jacobian $\hat{J}$ at the level of individual blocks in partial equilibrium — that is, taking the paths of aggregate inputs as given. We now show how these partial equilibrium objects are assembled into a general equilibrium solution, where the paths of all aggregate unknowns are determined simultaneously and consistently with market clearing. The key insight is that the behavioural expectations framework enters the general equilibrium solvent only through the partial equilibrium Jacobians of the blocks whose decision problems involve forward-looking expectations. This means we can replace FIRE Jacobians with their behavioural counterparts block-by-block — not only for the heterogeneous agents block but also for any simple block containing forward-looking equations (such as the Phillips curves, investment Euler equation, or asset pricing conditions) — and then solve for general equilibrium exactly as under FIRE.

A DSGE model can always be written as a root-finding problem in the space of length-T sequences of endogenous aggregate unknowns $\mathbf{U} \in R^{n\times T}$ and aggregate shocks $\mathbf{Z} \in R^{m\times T}$:

$$\mathbf{F}(\mathbf{U}, \mathbf{Z}) = \mathbf{0}$$

where the conditional expectations operator with respect to aggregates is suppressed as all uncertainty regarding the path of shocks is fully resolved and part of the information set at time zero. As shown in Auclert et al. (2021), a first order solution with certainty equivalence in the sequence-space can be obtained by linearising the above equilibrium equation in the aggregates around the steady state to obtain a mapping from (any) path of shocks resolved at time-zero and the path of aggregate unknowns:

$$\mathbf{F}'_U d\mathbf{U} + \mathbf{F}'_Z d\mathbf{Z} = 0$$

$$d\mathbf{U} = -\mathbf{H}_U^{-1}\mathbf{H}_Z d\mathbf{Z}$$

Where the $T \times T$ matrix

$$\mathbf{G} = -\mathbf{H}_U^{-1}\mathbf{H}_Z = \lim_{d\mathbf{Z}\to\mathbf{0}} \frac{d\mathbf{U}}{d\mathbf{Z}} \approx \frac{\partial \mathbf{U}}{\partial \mathbf{Z}}$$

[10] Asymptotic rationality, so that in the limit of the transition horizon going to infinity perfect foresight prevails, requires that the news-revision block-matrix $\mathbf{B}_h$ converges to an identity matrix:

$$\lim_{k\to\infty} \mathbf{B}_k = \mathbf{I},$$

which follows under our technical assumptions, particularly as shown in footnote (8).

is the *General Equilibrium Jacobian* collecting along the rows of each column $0 \leq s \leq T$ the linearised Impulse Response Functions (IRFs) of the aggregate unknown to a shock to the time s aggregate-shock process in a neighbourhood of the steady state. As shown in Auclert et al. (2021), exploiting the availability of a Directed Acyclic Graph (DAG) representation for the model, consisting of a partition into *target blocks* $\mathbf{H}(\mathbf{U}, \mathbf{Z})$ linking aggregate unknowns and the fundamental aggregate shocks via aggregate outputs of the heterogeneous agents block (i.e. *sinks* in the network vernacular), *simple blocks* $\mathbf{S}(\mathbf{X}, \mathbf{U}, \mathbf{Z}) = \mathbf{0}$ implicitly defining aggregate inputs as functions, and a *heterogeneous agents block* $\mathcal{Y} = \mathcal{K}(\mathbf{X}, \mathbf{Z})$ mapping inputs and shocks sequences into aggregate outputs. Given this structure, the General Equilibrium SSJ $\mathbf{G}$ can be efficiently computed by forward differentiation (FD) along the DAG, i.e. with the chain-rule:

$$\mathbf{G} = -[\mathbf{H}_{\mathcal{Y}}\tilde{\mathbf{J}}\mathbf{S}_X^{-1}\mathbf{S}_U]^{-1}[\mathbf{H}_{\mathcal{Y}}\tilde{\mathbf{J}}\mathbf{S}_X^{-1}\mathbf{S}_Z]$$

where the matrices $\mathbf{S}_X^{-1}\mathbf{S}_U$ and $\mathbf{S}_X^{-1}\mathbf{S}_Z$ consist of the partial equilibrium Jacobians of inputs with respect to the aggregate unknowns and shocks sequences evaluated at the steady-state.[11] The matrix $\tilde{\mathbf{J}}$ is the partial equilibrium Jacobian of the aggregate outputs with respect to the *true* aggregate inputs sequence $\mathbf{X}$. Crucially, $\tilde{\mathbf{J}}$ can be an arbitrary Jacobian, including non-FIRE Jacobians, as long as it is with respect to inputs only (as opposed to expectations thereon). We know how to compute such a Jacobian for non-FIRE cases from the previous section.

The important observation for our purposes is that the simple blocks $\mathbf{S}$ — which in our application include the forward-looking Phillips curves for price and wage inflation, the investment Euler equation, and the asset pricing conditions — also contain expectations of future aggregates. Under FIRE, the Jacobians of these blocks with respect to their inputs are computed under perfect foresight. Under our behavioural expectations framework, we replace the FIRE Jacobians of any forward-looking block, whether heterogeneous agent or simple, with their behavioural counterparts $\hat{J}$ constructed using the forecast mappings Λ_h from the previous section. In practice, this means that the matrices $\mathbf{S}_X^{-1}\mathbf{S}_U$ and $\mathbf{S}_X^{-1}\mathbf{S}_Z$ are also modified when the simple blocks that define them contain forward-looking equations subject to behavioural expectations. The general principle is straightforward: as long as a solvent $\mathbf{G}(\mathbf{J})$ applies under FIRE, there is a corresponding solvent $\mathbf{G}(\hat{\mathbf{J}})$ that applies under behavioural expectations, obtained by replacing the FIRE Jacobian $\mathbf{J}$ with its behavioural counterpart $\hat{\mathbf{J}}$ in every block where agents form expectations about future aggregates. This allows us to apply different behavioural parameters — different degrees of stickiness δ (or θ) and extrapolation γ — to different sectors of the model, as we do in our application where we distinguish between household and non-household (firm) expectations.

3 Application

As an application of the developed framework for flexible deviations from FIRE, we study the two-asset HANK model of Auclert et al. (2020) which emphasised the importance of departure from rational expectations in HANK models and demonstrated how to implement sticky expectations in the seqeuence space. We first replicate the model's steadystate and confirm we can replicate the macrodynamics (see Table 1 and Figure B.1). We then go on to estimate the models dynamic, behavioral and shock process parameters using a Bayesian full information maximum likelihood approach on quarterly US business cycle data including time series on inflation expectations from the Survey of Professional forecasters and Michigan survey.

[11] A similar expression including identity matrices is applied in cases where the unknowns or shocks figure explicitly in the target block rather than indirectly through the heterogeneous agents outputs:

$$\mathbf{G} = -[\mathbf{H}_{\mathcal{Y}}\tilde{\mathbf{J}}\mathbf{S}_X^{-1}\mathbf{S}_U + \mathbf{I}]^{-1}[\mathbf{H}_{\mathcal{Y}}\tilde{\mathbf{J}}\mathbf{S}_X^{-1}\mathbf{S}_Z + \mathbf{I}]$$

3.1 Two Asset Hank Model

The two-asset heterogeneous agent New Keynesian (HANK) model of Auclert et al. (2020), was designed to simultaneously match micro-level consumption dynamics and macro-level impulse responses. The benchmark model features rich heterogeneity on the household side that replicates key feature of the US distribution of disposable income and illiquid asset wealth, nominal rigidities in price and nominal wage setting, an aggregate investment side with time to build and adjustment costs, long-term government debt to replicate US maturity structures, and a quarterly calibration to match both key aggregates in the US economy and the micro-jump structure to intertemporal marginal propensities to consume (iMPCs). The next section outlines the main aspects of the model, steady state solution[12].

Heterogeneous households. Households differ in two main respects. First, there is ex-post heterogeneity in their exogenous idiosyncratic labour productivity levels $z_{i,t}$, assumed to follow a Markov process with transition matrix Π, and in their liquid asset positions $a_{i,t}$ as a consequence of uninsurable idiosyncratic risk and ad hoc borrowing constraints $a_{i,t} \geq -\bar{\phi}$ above the debt limit set to $\bar{\phi} = 0$. Second, ex-ante heterogeneity in terms of illiquid savings is assumed. In particular, it's assumed agents permanently lie in one of six non-overlapping groups g with measure μ_g, and such that $\sum \mu_g = 1$. Agents differ *between* groups g in terms of the average labour productivity level $\bar{e}_g$, the discount factor β_g, and steady state illiquid asset positions $\bar{a}_g$. Agents *within* the same group g instead are identical in the above respect, and only differ ex-post along the idiosyncratic productivity and liquid position dimensions. As shown in Auclert et al. (2020), persistent or ex-ante heterogeneity in the above respects can be exploited to obtain a two-asset structure in a HANK model that replicates the empirical inequality in the illiquid asset distribution yet still simple enough to be computationally inexpensive, which is key for the estimation procedures in our applications when requiring re-computation of the model Jacobians. In each period, each group-g household with instantaneous preferences $f(c, n) = u(c) - v(n)$ separable over consumption and labour observes its productivity realisation $\bar{e}_g z_t$ and given the tax rate τ_t, its real disposable income $(1-\tau_t)w_t N_t z_t \bar{e}_g z_t$ from supplying labour $n_t = N_t$ *inelastically* to meet the demand from labour unions, under the assumption that these allocate labour demand uniformly across households, the value of its liquid asset portfolio $(1 + r_t^l)l_{i,t-1}$, and the amount of assets converted from the illiquid portfolio a to disposable resources at the beginning of the period $d_{g,t}$. The household chooses consumption and its end of period liquid asset position to maximise the expected present value of its consumption and labour supply streams, subject to the budget and borrowing constraints. Recursively, this amounts to solving the recursive Bellman equation:

$$V_{g,t}(l, a, z; \mathbf{X}_t) = u(c_t) - v(N_t) + \beta_g E_{z'|z} V_{t+1}(l', a', z'; \mathbf{X}_{t+1})$$
$$c + l' = (1 - \tau_t) w_t N_t \bar{e}_g z_t + (1 + r_t^l) l_t + d_{g,t}$$
$$l' \geq 0$$
$$a' = (1 + r_t^a) a - d_{g,t}$$

where $\mathbf{X}_t = [w_t, r_t^l, \tau_t, N_t, d_{g,t}]'$ is the relevant aggregate state vector (for group-g households) at time t, and the distribution policy from the illiquid portfolio to disposable resources $d_{g,t}$ is modelled by a simple rule, for $\chi \to 0^+$:

$$d_{g,t} = \frac{r_{ss}^a}{1 + r_{ss}^a}(1 + r_t^a)a + \chi[(1 + r_t^a)a - (1 + r_{ss}^a)\bar{a}_g].$$

From the law of motion for a, at the non-stochastic steady state the value of the distribution policy is such to consume the return on the stationary illiquid portfolio (i.e. the annuity value of gross returns to the asset):

$$d_g = r_{ss}^a \bar{a},$$

[12]For an overview of the model in greater detail, we redirect the reader to the original paper Auclert et al. (2020) and Appendix B.1.

which combined with the simple rule for $d_{g,t}$ implies:

$$a = \bar{a}_g.$$

The above model then implies that agents save when the value of the illiquid portfolio, via the stock market, is low and dissave when high (relative to the steady state), and the illiquid asset holdings converge at the steady state to the group-specific average a_g.

Mutual funds and financial investment. Financial investment is carried out by a perfectly competitive mutual fund, who issues liquid and illiquid account varieties to households. Its balance sheet consists of liabilities to households, in the form of liquid and illiquid assets accounts held by households, while assets include stocks (with their aggregate supply normalised to unary mass in equilibrium) in the capital-building and intermediate firm commanding a shareholder value consisting of the sum of end of period dividend and price, government bonds, and central-bank reserves ex-post set in zero net supply. The intermediary faces linear intermediation costs on the liquid accounts, which implies that the liquid asset commands in equilibrium a rate of return strictly lower than the illiquid one. The competitive investment fund's optimisation problem is hence to choose to investment through each asset to maximise expected profits, subject to the balance sheet constraint.[13] This results in a set of asset-pricing conditions for the economy and a fisher equation, from the first order condition for reserves, describing the pass-throughs from monetary policy shocks via the ex ante rate to asset prices.

Firms. On the production side, the model features nominal rigidities both in price-setting by intermediate producers, with a Cobb-Douglas technology including labour and capital stock as production factors, and in wage setting by a representative union that sells households labour services to a competitive labour packer. The set-up results in a combination of sticky-prices and sticky-wages known to capture well the sluggish adjustment of the price level and feedback between price and wage inflation.. The real investment sector consists of capital-building firms, and includes time-to-build constraints and Rotemberg-style quadratic adjustment costs, which generate realistic and sluggish investment dynamics around the steady state. In particular, the frictional set-up implies investment takes time to respond to time-zero monetary policy shocks even under FIRE and as such that the income effect it exerts on households' budgets is sustained over time serving as a source of amplification. As Auclert et al. (2020) point out, hump-shaped investment and income dynamics are not enough on their own to match hump-shaped monetary policy shocks, and deviations from FIRE are needed.

Fiscal and Monetary Policy. The government issues long-term debt with a maturity structure matching US Treasury data, allowing bond prices to respond meaningfully to monetary policy shocks. It collects tax revenue according to a simple fiscal rule where the tax rate slowly adjust to bring real government debt levels back to it's steady state value. Monetary policy follows a standard Taylor rule responding to inflation and output.

Calibration The quarterly calibration of the model replicates the strategy in Auclert et al. (2020). It targets both aggregate moments of the US economy, particularly aggregate wealth, rate of returns, and bond-maturity structures – and via between-group permanent differences in illiquid asset holdings and discount factors the micro-structure of consumption responses to income shocks (iMPCs) and illiquid asset distribution, positioning the model to address the dual challenge of matching micro-jumps and macro-humps simultaneously. We replicate the steady state of the model which is detailed in Table 1.

3.2 Business Cycle Estimation

Our next step is to estimate the model's dynamic parameters on US business cycle data that includes inflation expectations for both professional and households as observables. Specifically we conduct

[13] I.e. equivalent by construction here to asset market clearing

Table 1: Model Calibration and Estimation

Panel A: Calibrated Parameters

Parameter		Value	Parameter		Value
ν	EIS	1	α	Capital share	0.24
ς	Frisch	0.5	δ_K	Depreciation (p.a.)	0.053
r	Real interest rate (p.a.)	0.050	μ_p	Price markup	1.06
K/Y	Capital to GDP (p.a.)	2.23	L/Y	Liquid assets to GDP (p.a.)	0.23
ξ	Intermediation spread (p.a.)	0.065	G/Y	Spending-to-GDP	0.16
qB/Y	Gov. bonds to GDP (p.a.)	0.42	$\frac{1+r}{1+r-\delta}$	Debt maturity (a.)	5
ψ	Tax response to debt (p.a.)	0.1	ϕ_π	Taylor rule coefficient	1.5

Panel B: Household Heterogeneity

Group g	1	2	3	4	5	6
Description	Bottom 50%	Next 20%	Next 10%	Next 10%	Next 5%	Top 5%
Population share (μ_g)	0.500	0.200	0.100	0.100	0.050	0.050
Illiquid asset share	2.7%	7.0%	7.0%	13.0%	12.2%	58.0%
Labor income share	26.7%	18.3%	10.8%	14.4%	11.0%	18.8%
Discount factor β_g (p.a.)	0.905	0.919	0.933	0.946	0.950	0.975

a full information maximum likelihood estimation of these parameters using nine quarterly macroeconomic time series over the period 1978-2018 for: GDP, consumption, business investment, hours worked, real wages, value added inflation, FED's fund rate, professional forecaster inflation expectations and household inflation expectations. The first seven series are those used in Smets & Wouters (2007) and Auclert et al. (2020), following the transformations of Smets & Wouters (2007), described in the appendix (see Figure B.2). The inflation expetation series are the quarterly averages of survey respondents one year ahead expectations, linearly de-trended in line with actual inflation outturns. In order to estimate the model we include seven exogenous shock processes (following Auclert et al. (2020)) and two measurement error terms for the two inflation series with the prior mean of the measurement error set in line with the average of the boot-strapped quarterly standard deviation of each surveys quarterly mean. The seven exogenous processes are all AR1 with the wage and price inflation mark-up series further including an MA1 term.[14]

The addition of these exogenous processes adds sixteen shock process parameters to the model to be estimated. We further estimate four dynamic parameters related to the real and nominal rigidities in the model: investment adjustment cost parameter ψ, Taylor rule inertia ρ_m, price and wage update probabilities ζ. For all these parameters we borrow the priors form Auclert et al. (2020) which in turn are largely borrowed from Smets & Wouters (2007). It's worth noting that these parameters were estimated via IRF matching in Auclert et al. (2020) before estimating the shock parameters as a second step on the business cycle data.

Finally we introduce the estimation of four behavioral parameters $\{\theta_{hh}, \theta_{prof}, \gamma_{hh}, \gamma_{prof}\}$ that estimate households and firms information stickiness and tendency to extrapolate as outlined in the previous section.[15] [16] As in Auclert et al. (2020), these information frictions are applied equally to all household inputs $\{w, r^l, \tau, N, d_g\}$ and for firms and financial markets they are applied to the relevant inputs to their forward looking equations for value added price inflation, investments and dividends. In order to use the inflation expectations series to help pin down the behavioral parameters we included separate equations for four quarter ahead inflation forecasts for households and firms, that incorporate inattentiveness and extrapolation with respect to current and future inflation.

Table 2 summarises the parameters to be estimated, their priors, and the modes and confidence intervals calculated using a Randow Walk Metropolis Hastings Chain of length 120,000. We also

[14]The seven shock processes include: a discount factor shock, monetary policy shock, government spending shock, TFP shock, risk premium shock, price and wage markup shock.

[15]For γ_{hh} and γ_{prof} in practice we decay this parameter at a quarterly decay rate of 0.86 beginning from quarter 20 after a shock. This in practice limits extrapolation to 10 years, beginning the decay from 5 years, and was introduced having observed the tendency of extrapolation to produce long-run cycles when applied on both the household and firm side. This did not noticeably effect the estimation of the parameters but did significantly extend the time over which IRF's converged back to zero which was of practical cost to the estimation process.

[16]The stickiness parameter θ can be mapped to the δ_x form of the previous section by the transformation: $\theta = 1 - \frac{1}{1+\delta_x}$.

report the modal estimates where we choose not to estimate the extrapolation terms as a benchmark to compare the benefit of including this term beyond the inattentiveness/sticky terms Prior to starting the chain we estimate the mode using the Nelder-Mead algorithm up to a max of 4800 iterations. We obtain the starting mode and parameter covariance-matrix by a further search around the Nelder-Mead mode using the L-BFGS-B algorithm. We report the individual parameter chains and Geweke convergence test in appendix Figures B.3 and B.4.

Turning to the reported values in Table 2, in line with the existing literature our procedure estimates highly persistent shock processes for both the sticky and sticky and extrapolative case. This is somewhat disappointing as the extra potential for endogenous persistence brought about by the extrapolative expectations channel might have been expected to substitute for exogenous shock persistence, as suggested by the learning literature (e.g. Slobodyan & Wouters (2012)), who find that replacing rational expectations with adaptive learning in the Smets & Wouters (2007) model lowers estimated persistence of price and wage markup shocks. In our setting, despite the additional endogenous propagation from extrapolation, the data continues to demand highly persistent exogenous driving processes, particularly for the discount factor and TFP. Comparing the sticky only to sticky and extrapolation case, a notable exception is the risk premium shock persistence, which falls substantially compared to the no-extrapolation case. This is further accompanied by a dramatic reduction in the process standard deviation, suggesting that the extrapolative channel partially substitutes for the exogenous risk premium process in generating the requisite volatility in asset prices and investment. Similarly, the price markup shock persistence drops markedly when extrapolation is included, alongside a substantial reduction in its shock standard deviation. Indicating that this extra behavioral channel absorbs some of the role previously played by exogenous cost-push disturbances in fitting inflation dynamics.

Turning to the dynamic parameters, the inclusion of extrapolation has meaningful effects on several key structural estimates. The investment adjustment cost parameter ψ falls consistent with the intuition that extrapolative expectations generate additional endogenous amplification in investment dynamics, reducing the need for large adjustment costs to match the hump-shaped investment responses observed in the data. Taylor rule inertia ρ_m in the extrapolative case, suggesting that once expectations and inflation overreact to current conditions, the estimation attributes greater smoothing to the monetary authority. The Calvo price stickiness parameter ζ_p implies an average price durations of roughly 3 quarters, in the neighbourhood of the Smets & Wouters (2007) estimates. Wage stickiness ζ_w is very high at in both specifications, implying near-complete wage rigidity at the quarterly frequency.

The behavioural parameter estimates represent a key contribution of our analysis. On the firm or non-household side, our estimates imply that professional forecasters and firms update their information sets on average every two quarters. This is broadly in line with the empirical estimates of Coibion & Gorodnichenko (2015), who find information rigidity coefficients for professional forecasters in the range of 0.50 to 0.7. Notably, when extrapolation is turned off, the estimated stickiness on the firm side rises substantially implying average durations between information updates of around six quarters. Once agents are allowed to overreact to current conditions less stickiness is needed to match the same patterns in expectations data. The negative sign and magnitude of γ_{prof} is empirically close to the overreaction to current conditions documented by Kohlhas & Walther (2021).

On the household side, the estimated stickiness parameter is extremely high and tightly estimated, implying that households update their information sets very infrequently—on average. This is substantially higher than the micro-evidence would suggest and higher even than the non-expectations data based IRF matching based estimate used in Auclert et al. (2020). While the high stickiness successfully generates the requisite macro humps and delayed responses in the model's impulse response functions, it is difficult to reconcile with survey evidence suggesting households update expectations more frequently. The household extrapolation parameter γ_{hh} is estimated to be negative and statistically different from zero indicating overreaction to current conditions also on the household side, though with somewhat less precision than for firms. Finally, the overall model fit, as measured by the mode of the log-posterior, improves substantially relative to the no-extrapolation case with a likelihood ratio test statistic of 213 with two degreees of freedom decisively rejecting the null hypothesis

$(\gamma_{hh}, \gamma_{prof} = 0)$.

Table 2: Parameter Estimates

Parameter	Prior	Baseline					No Extrap.
		Mode	Mean	Std	5%	95%	Mode
Shock Processes							
ρ_C	Beta(0.5,0.2)	0.962	0.956	0.015	0.931	0.979	0.969
ρ_r	Beta(0.5,0.2)	0.210	0.254	0.089	0.114	0.405	0.452
ρ_g	Beta(0.5,0.2)	0.895	0.890	0.040	0.821	0.953	0.880
ρ_z	Beta(0.5,0.2)	0.987	0.986	0.003	0.981	0.990	0.986
ρ_{rp}	Beta(0.5,0.2)	0.758	0.742	0.040	0.676	0.810	0.588
ρ_p	Beta(0.5,0.2)	0.338	0.344	0.126	0.144	0.556	0.745
ρ_w	Beta(0.5,0.2)	0.987	0.986	0.003	0.980	0.991	0.992
σ_C	Inv-Gamma(0.1,2)	0.458	0.483	0.065	0.385	0.597	0.434
σ_r	Inv-Gamma(0.1,2)	0.269	0.274	0.018	0.246	0.305	0.282
σ_g	Inv-Gamma(0.1,2)	0.298	0.301	0.016	0.275	0.330	0.297
σ_z	Inv-Gamma(0.1,2)	0.846	0.867	0.060	0.773	0.972	1.242
σ_{rp}	Inv-Gamma(0.1,2)	3.159	3.982	1.282	2.133	6.299	11.990
σ_p	Inv-Gamma(0.1,2)	0.161	0.185	0.034	0.133	0.243	0.529
σ_w	Inv-Gamma(0.1,2)	0.272	0.273	0.022	0.238	0.311	0.281
θ_p	Beta(0.5,0.2)	0.772	0.758	0.097	0.595	0.911	0.960
θ_w	Beta(0.5,0.2)	0.976	0.975	0.006	0.964	0.983	0.983
Dynamic							
ψ	Normal(10,4)	7.251	8.033	1.394	5.991	10.679	9.530
ρ_m	Beta(0.5,0.2)	0.793	0.769	0.044	0.695	0.841	0.639
ζ_p	Beta(0.7,0.2)	0.655	0.638	0.035	0.577	0.697	0.492
ζ_w	Beta(0.7,0.2)	0.992	0.992	0.001	0.992	0.993	0.992
Behavioral							
θ_{hh}	Beta(0.5,0.2)	0.984	0.984	0.001	0.982	0.985	0.986
θ_{prof}	Beta(0.5,0.2)	0.500	0.519	0.044	0.440	0.588	0.833
γ_{hh}	Normal(0.0,0.3)	-0.134	-0.151	0.089	-0.308	-0.014	0.000
γ_{prof}	Normal(0.0,0.3)	-0.207	-0.194	0.029	-0.245	-0.150	0.000
Mode log-posterior				-379			-485

Note: Table reports the parameter estimates and confidence bands after a 120,000 long draw Random Walk Metropolis Hastings chain with acceptance ratio of 25.3%. We discard the first 30,000 draws and report convergence statistics in the appendix. We also report the estimated mode for a version of the model with extrapolation parameters set to zero.

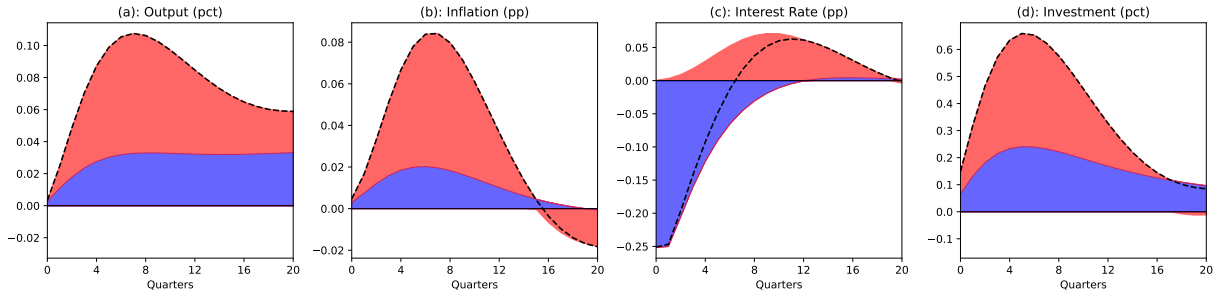
3.3 Role of extrapolation

We now further examine the properties of including extrapolation in the estimated model, focusing on three dimensions: the amplification of impulse responses, the dynamic pattern of forecast errors, and the model's forecasting performance.

Amplification through extrapolation. Figure 1 isolates the contribution of extrapolation to the transmission of monetary policy shocks. We do so by re-solving the model at the estimated modal parameters from Table 2 but setting the extrapolation parameters γ_{hh} and γ_{prof} to zero for both households and firms. The blue lines in Figure 1 trace out the impulse responses under this counterfactual — a model with the same degree of estimated stickiness but no extrapolation — while the red shaded area represents the additional effect attributable to extrapolation.

The mechanism is intuitive. Following a expansionary monetary policy shock, output and inflation increase on impact. Under pure sticky expectations, only a fraction $(1 - \theta)$ of agents update their information set each period, and those who do update acquire the correct (perfect foresight) path. The sluggish diffusion of information through the population generates a hump-shaped response, but because agents who update do so without bias, the peak response is substantially attenuated relative to a world where all agents immediately perceive the shock. Extrapolation fundamentally changes this: when agents do update, they not only incorporate news about the future path of aggregates but also extrapolate from the current state of the economy ($\gamma < 0$ implying overreaction to current

Figure 1: Extrapolation contribution to a monetary policy shock



Note: This figure illustrates the contribution of extrapolation in the estimated model by re-solving the model with $\gamma_0 = \gamma_{prof} = 0$ for both firms and households, otherwise setting all parameters at the posterior mode reported in Table 2. Blue lines show the impulse responses without extrapolation; the red shaded area represents the additional contribution of extrapolation. The shock is a 25bp contractionary monetary policy shock. For other shocks see Appendix B.4.2

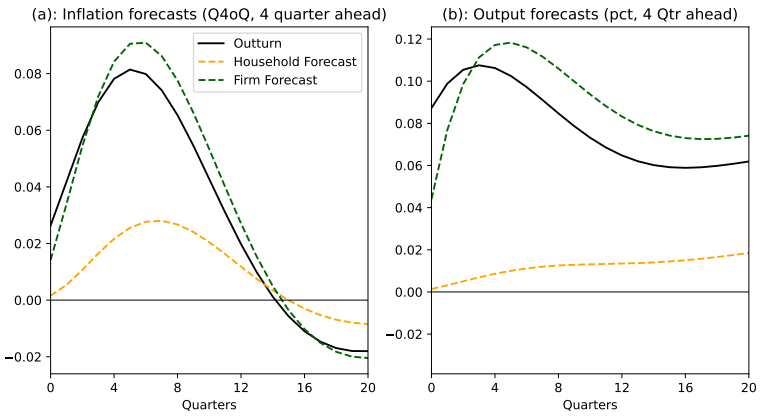
conditions). In the aftermath of an expansionary shock, the increase in output and inflation at the time of updating causes agents to form forecasts that are *more optimistic*. This pro-cyclical bias in the forecasts of updating agents amplifies the expansionary impulse, generating a substantially larger peak response while the stickiness parameter continues to delay when that peak occurs.

As highlighted in the related work of Albuquerque et al. (2025), this interaction between extrapolation and stickiness is synergistic: extrapolation provides amplification while stickiness provides persistence, and together they produce large hump-shaped responses that neither channel can generate alone. The quantitative effects in this instance are considerable. Extrapolation approximately triples the peak response of investment and quadruples the peak inflation response relative to the sticky-only counterfactual. The mechanism for the interest rate response differs because the Taylor rule sets the nominal rate based on contemporaneous outturns of inflation and output rather than on expectations. The amplified decline in inflation and output caused by extrapolative expectations therefore feeds back into a more aggressive monetary easing in the periods following the initial shock, with the interest rate returning to and overshooting its steady state value more rapidly than in the sticky-only case.

Dynamic pattern of forecast errors. Figure 2 plots the model-implied four-quarter-ahead forecasts of inflation and the log-level deviation of output from steady state for both firms and households, alongside the actual model outturns, in response to the same monetary policy shock. On the firm side, the forecasts display the characteristic pattern of initial underreaction followed by subsequent overreaction documented empirically by Angeletos et al. (2021). The intuition follows directly from the forecast error law of motion derived in Section 2. In the periods immediately following the shock, the majority of firms have not yet updated their information sets ($\theta_{prof} = 0.50$ implies approximately half update each quarter), so the average forecast lags behind the true path — generating underreaction. As more firms update over subsequent quarters, those who do update extrapolate from the depressed state of the economy at the time of their update, forming forecasts that overshoot the actual path of the variable. The combination of progressive updating and pro-cyclical extrapolation thus generates a sign reversal in the forecast error: initially positive (underreaction, forecast above outturn for a declining variable) and subsequently negative (overreaction, forecast below outturn). This is precisely the "dynamic overshooting" pattern that Angeletos et al. (2021) argue requires the combination of dispersed information and overextrapolation, and which our framework captures parsimoniously through δ_x/θ_x and γ_x.

On the household side, this mechanism is substantially muted. The estimated household stickiness of $\theta = 0.984$ implies that only around 1.6% of households update their information set each quarter. This extreme inattention means the household forecast responds very slowly to the shock, with the output forecast in particular exhibiting a protracted and gradual adjustment. While the estimated household extrapolation parameter ($\gamma_{hh} = -0.134$) would generate overreaction among those house-

Figure 2: Firm and household forecasts



Note: Figure plots the four-quarter-ahead forecasts and actual model outturns for inflation and the log-level deviation of output from steady state, for both firms (top row) and households (bottom row). Responses are to the same 25bp contractionary monetary policy shock as in Figure 1, at the posterior modal parameter estimates from Table 2.

holds who do update, the tiny fraction updating each period means this channel has limited scope to operate in aggregate. The household forecast therefore remains dominated by the prior, with the sign-reversal pattern largely absent over the horizons plotted. As noted earlier, this extreme household stickiness is difficult to reconcile with the micro-evidence suggesting more frequent updating, and points to a potential tension in the model that merits further investigation.

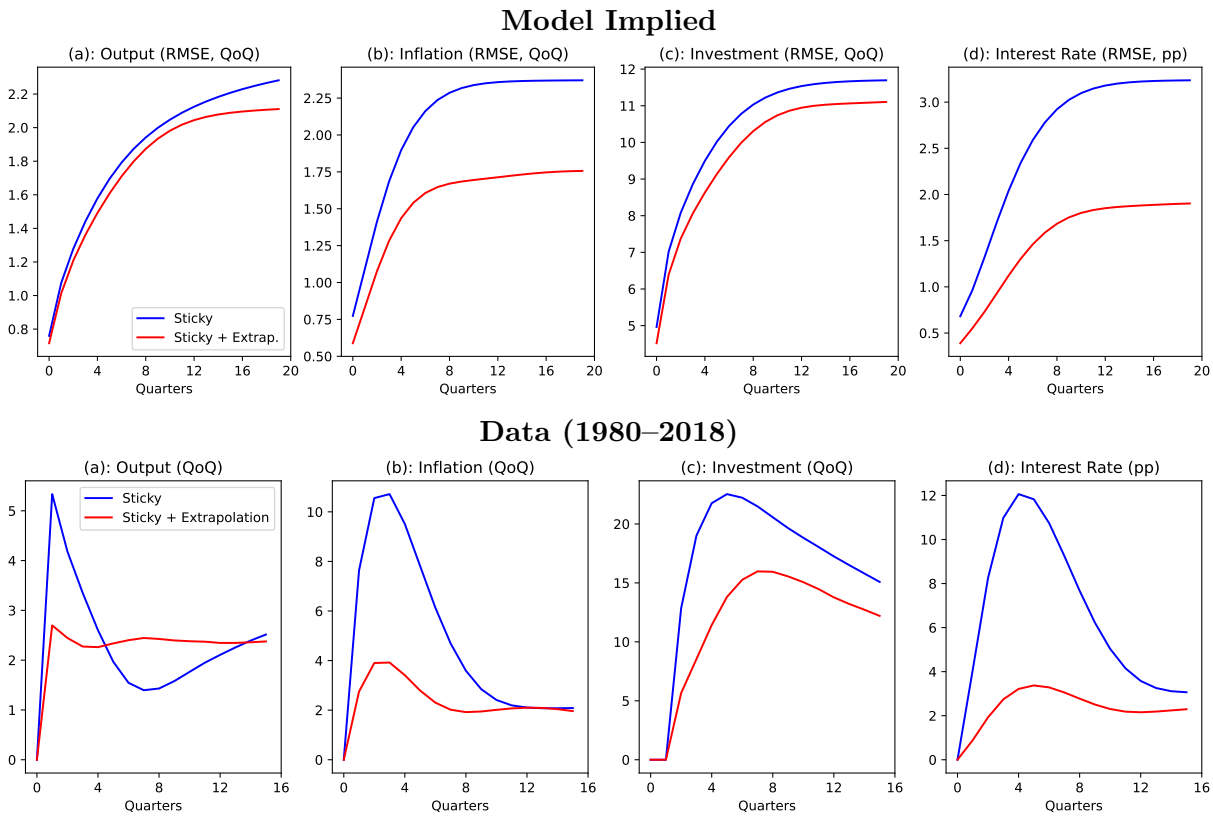
Forecasting performance. A natural question is whether the improved in-sample fit from including extrapolation — reflected in the 106 log-posterior point improvement documented in Table 2 — translates into better forecasting properties. Figure 3 addresses this by comparing the root mean square forecast errors (RMSEs) of the baseline model (with extrapolation) against the restricted model (sticky expectations only) across forecast horizons and observable series. The top panel reports model-implied RMSEs, computed by simulating the model at the estimated shock processes from Table 2 and evaluating the forecast errors at each horizon. The bottom panel reports RMSEs computed against actual US data outturns over the 1980–2018 sample period, using the model's forecasts at the modal parameter estimates.

Both panels tell a consistent story. The baseline model with extrapolation delivers lower RMSEs than the sticky-only model, with the gains concentrated in the inflation and interest rate series and at medium-to-longer forecast horizons. This pattern is economically intuitive. At very short horizons, the two models produce similar forecasts because the extrapolative channel has had little time to operate and the dominant source of forecast variation is the arrival of new shocks. At longer horizons, however, the sticky-only model's forecasts are too sluggish — they fail to capture the amplified dynamics that extrapolation generates in the data. In effect, the sticky-only model must rely on larger and more persistent exogenous shocks (as reflected in the higher σ_{rp} and σ_p estimates in Table 2) to fit the same data, and these larger shocks do not translate into better out-of-sample forecasting because they represent overfitting to in-sample volatility through the exogenous processes rather than capturing the endogenous propagation mechanism. By contrast, the model with extrapolation achieves a better fit with smaller exogenous shocks by attributing more of the observed dynamics to the endogenous amplification from behavioural expectations, which in turn produces more accurate forecasts at the horizons where this propagation mechanism matters most.

4 Conclusion

This paper proposes a flexible and standardised approach to departing from full information rational expectations (FIRE) in DSGE models, and particularly heterogeneous agent New Keynesian (HANK)

Figure 3: Model forecast errors



Note: Figure plots root mean square forecast errors across horizons for selected observable series. The top panel reports model-implied RMSEs based on the estimated shock processes from Table 2. The bottom panel reports in-sample RMSEs against US data outturns over 1980–2018, evaluated at the posterior modal parameter estimates. Blue bars correspond to the model without extrapolation ($\gamma_0 = \gamma_{prof} = 0$); red bars correspond to the baseline model with extrapolation.

models, using the sequence space framework. Building on the empirical evidence from Kohlhas & Walther (2021), we implement a reduced-form behavioural expectations process such that agents simultaneously overreact to current outcomes and underreact to news. The reduced-form approach can be microfounded via limited attention mechanisms but can nests various expectation models, while maintaining a clear mapping to empirical evidence and remaining agnostic on the sources of the informational and belief frictions. We apply the developed framework and sequence-space solvent to the solution full-information Bayesian estimation of a medium-scale two-asset HANK model featuring rich and quantitatively realistic heterogeneity and macro frictions. We estimate that both households and firms simultaneously extrapolate to outcomes today and under react to news about tomorrow, with households significantly less reactive to news about tomorrow than firms. We show that in the model, extrapolation leads to amplification of impulse responses to aggregate shocks, characteristic initial under reaction and subsequent over-reaction of forecasts of firms, and improved forecasting performance.

References

Adam, K. & Marcet, A. (2011), 'Internal rationality, imperfect market knowledge and asset prices', *Journal of Economic Theory* **146**(3), 1224–1252.

Adam, K., Pfäuti, O. & Reinelt, T. (2024), 'Subjective housing price expectations, falling natural rates, and the optimal inflation target', *Journal of Monetary Economics* p. 103647.

Albuquerque, D., Lazarowicz, T. & Lenney, J. (2025), 'Monetary transmission through the housing sector'. Working paper.

Angeletos, G.-M., Guerreiro, J. & Zhang, D. R. (2025), From rank to hank, without fire, Technical report, National Bureau of Economic Research.

Angeletos, G.-M., Huo, Z. & Sastry, K. A. (2021), 'Imperfect macroeconomic expectations: Evidence and theory', *NBER Macroeconomics Annual* **35**(1), 1–86.

Auclert, A., Bardóczy, B., Rognlie, M. & Straub, L. (2021), 'Using the sequence-space jacobian to solve and estimate heterogeneous-agent models', *Econometrica* **89**(5), 2375–2408.

Auclert, A., Rognlie, M. & Straub, L. (2020), Micro jumps, macro humps: Monetary policy and business cycles in an estimated hank model, Technical report, National Bureau of Economic Research.

Bardóczy, B. & Guerreiro, J. (2024), 'Unemployment insurance in macroeconomic stabilization with imperfect expectations', *Manuscript, April* **5**.

Bordalo, P., Gennaioli, N., Ma, Y. & Shleifer, A. (2020), 'Overreaction in macroeconomic expectations', *American Economic Review* **110**(9), 2748–2782.

Bordalo, P., Gennaioli, N. & Shleifer, A. (2018), 'Diagnostic expectations and credit cycles', *Journal of Finance* **73**(1), 199–227.

Coibion, O. & Gorodnichenko, Y. (2015), 'Information rigidity and the expectations formation process: A simple framework and new facts', *American Economic Review* **105**(8), 2644–2678.

Del Negro, M. & Eusepi, S. (2011), 'Fitting observed inflation expectations', *Journal of Economic Dynamics and control* **35**(12), 2105–2131.

Del Negro, M. & Schorfheide, F. (2013), Dsge model-based forecasting, *in* 'Handbook of economic forecasting', Vol. 2, Elsevier, pp. 57–140.

Kohlhas, A. N. & Walther, A. (2021), 'Asymmetric attention', *American Economic Review* **111**(9), 2879–2925.

Mankiw, N. G. & Reis, R. (2003), 'Sticky information: A model of monetary nonneutrality and structural slumps', *Knowledge, information, and expectations in modern macroeconomics: In honor of Edmund S. Phelps* pp. 64–86.

Romer, C. D. & Romer, D. H. (2004), 'A new measure of monetary shocks: Derivation and implications', *American economic review* **94**(4), 1055–1084.

Slobodyan, S. & Wouters, R. (2012), 'Learning in an estimated medium-scale dsge model', *Journal of Economic Dynamics and Control* **36**(1), 26–46.
URL: *https://www.sciencedirect.com/science/article/pii/S0165188911001217*

Smets, F. & Wouters, R. (2007), 'Shocks and frictions in us business cycles: A bayesian dsge approach', *American economic review* **97**(3), 586–606.

A Further Methodological Details

A.1 Law of Motion for Behavioural Forecasts around the Steady State

Law of Motion. Here it is shown how the average forecast for the aggregate input at a time k in the cross-section of agents evolves over time around the steady state path. Recall that the forecast formed at time t on an aggregate input k periods ahead is given by:

$$\bar{f}_t x_{t+k} = c + \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1} x_{t+k} + x_{t+k} - \gamma x_t\right)$$

Where we have suppressed the expectations operator E_t^* as, by virtue of the Sequence Space solution concept, there is no aggregate uncertainty and the sequences in the aggregate are part of the full information set at time zero. Performing a linear perturbation around the steady state, where $x_r = x_{ss} + dx_r$:

$$\bar{f}_t(x_{ss} + dx_{t+k}) = c + \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1}(x_{ss} + dx_{t+k}) + x_{ss} + dx_{t+k} - \gamma(x_{ss} + dx_t)\right)$$

Using the fact that $f_t x_{ss} = f_{ss} x_{ss}$, together with the *no bias assumption* in the steady state pinning down the value of the constant $c = \frac{\gamma}{1+\delta} x_{ss}$:

$$x_{ss} + f_t dx_{t+k} = x_{ss} + \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1} dx_{t+k} + dx_{t+k} - \gamma dx_t\right)$$

$$f_t dx_{t+k} = \frac{1}{1+\delta}\left(\delta \bar{f}_{t-1} dx_{t+k} + dx_{t+k} - \gamma dx_t\right) \tag{A.1}$$

The above equation yields a recursive law of motion or difference equation (in the household's information sets) for how the forecast of the input in period k period ahead evolves as a function of past expectations $\bar{f}_{t-1} x_{t+k}$ formed in period $t-1$ on the same input ($k+1$ periods ahead), the full information set, and extrapolation from current events. Solving the difference equation backwards together with the assumed boundary condition $\bar{f}_{-1} dx_{t+k} = \bar{f}_{ss} dx_{t+k} = 0$ yields Equation 3 in the main text.

A.2 Technical Assumptions for Internal Rationality

Our framework for behavioural expectations in DSGE and HANK based on the Kohlhas-Walther mapping rests on a set of technical assumptions in order to ensure that agents are *internally rational* in the sense of Adam and Marcet (2011). These are:

1. No Bias along the Steady State path

$$c := \frac{1}{1+\delta}\int a_j P(dj) = \frac{1}{1+\delta}\gamma x_{ss}$$

$$\int a_j P(dj) = \gamma x_{ss}$$

2. A Law of Iterated Expectations holds for the expectations operator. That is, $\forall t, s$ with $s = t - r, r \geq 0$ and where we assume that the information set lies in a natural filtration, so that agents do not expect or forecast their own forecast errors:

$$\bar{f}_s[\bar{f}_t x_{t+k}] = \bar{f}_s[x_{t+k} - \epsilon_{t+k|t}] = \bar{f}_s x_{t+k} = \bar{f}_s x_{s+r+k}$$

That is, future forecast errors are orthogonal to the subjective information sets in force at the time expectations are formed. By the *Projection Theorem*, agents forecast under the assumed behavioural expectations are thus also efficient in the sense that they achieve the minimum MSE. As we discuss in the next section this is is critical in correctly pinning down the outlines heterogeneous agents block Jacobian, in that they imply that forecast updates between periods hit as time zero shocks.

3. For large enough T, agents expect the forecast variable to return to the steady state Bardóczy & Guerreiro (2024):

$$\exists T \in N : \forall t \leq T \quad \bar{f}_t x_T = x_T = x_{ss} = \bar{f}_{ss}\bar{x}_{ss}$$

4. News revision coefficient $\delta > 0$, which is sufficient to guarantee *stability* of the law of motion for the forecast in the sequence space around the steady state forecast.

A.3 The Heterogeneous Block Jacobian for the Behavioural Mapping $\mathcal{K}_t$

This section derives the outlined Jacobian of the heterogeneous agents block under Kohlhas-Walther deviations from FIRE as a function of the FIRE Jacobian from first principles, exploiting the procedure in Bardoczy and Guerreiro (2024) that extends the construction of the sequence space equilibrium Jacobian in Auclert et al. (2021) to environments in which agents' expectations on aggregate inputs do not necessarily coincide with the actual or true aggregate input sequence.

Consider any i-th scalar endogenous aggregate output $Y_{i,t} \in R_+$ of the heterogeneous agents models. In the DAG formulation of the sequence-space equilibrium this satisfies (Auclert et al. 2021; Bardoczy and Guerreiro, 2024):

$$Y_{i,t} = h(\mathbf{X})$$

Where $h : \mathcal{X} \to R_+$ is the *heterogeneous agents block* mapping the set of length-T sequences of n_i aggregate inputs $\mathcal{X} \subset R^{n_i \times T}$ to the household decision problem into the reals. The set of aggregate inputs is most generally taken to include both the true equilibrium aggregate input sequences $\mathbf{X}_s$ and the expectations heterogeneous agents form on them, a family $\{\mathbf{X}_s^{e,t}\}_{t=1,...,T}$, where t indicates the time at which expectations are formed and s the timing of the forecast aggregate input for $s \in 1, ..., T$. Making dependence on expectations explicit, the heterogeneous agents mapping with time-varying non-rational expectations is:

$$Y_{i,t} = h(\mathbf{X}, \mathbf{X}^{e,0}, ..., \mathbf{X}^{e,t}, ..., \mathbf{X}^{e,T}) = h(\mathbf{X}, \mathbf{X}^{e,0}, ..., \mathbf{X}^{e,t}).$$

The second equality embeds an assumption that, at time t, only the expectations prevailing up to that time matter to the agents decision-making. In other words, we assume that the endogenous aggregate output at any time t does not depend on or respond on expectations that will be formed in the future, that is on $\mathbf{X}^{e,\tau}$ for $\tau > t$. This is reasonable as changes in the endogenous aggregate output at any time t depend solely on the optimal policies formed at time t and the beginning of period distribution of agents over the idiosyncratic state space, in turn a function of past optimal policies – our assumption then states that optimal policies at any time cannot make use of information that is not part of the agents information sets at such time. Note that this assumption simplifies/reduces the dimension of the problem but is equivalent to the analysis in Bardoczy and Guerreiro. Finally, under a FIRE assumption such that $\forall t \quad \mathbf{X} = \mathbf{X}^{e,t}$, it is clear that the mapping collapses to the standard one in Auclert et al. (2021).

Consider for simplicity and without loss of generality a case with one aggregate input. At a first order perturbation around the non-stochastic steady state path in the sequence space, $\forall t = 1, ..., T$:

$$dY_{i,t} = \frac{\partial Y_{i,t}}{\partial X_0} dX_0 + ... + \frac{\partial Y_{i,t}}{\partial X_T} dX_T + \sum_{\tau=0}^{t} \left[\frac{\partial Y_{i,t}}{\partial X_0^{e,\tau}} dX_0^{e,\tau} + ... + \frac{\partial Y_{i,t}}{\partial X_T^{e,\tau}} dX_T^{e,\tau}. \right]$$

This is the key object we work with for constructing the Jacobian of interest. The second assumption we make is that there is a time-varying linear mapping from the sequence of aggregate equilibrium inputs $\mathbf{X}$ to the agents expectations/forecasts $X^{e,t}$ prevailing at the time. That is, we assume that for all t there is some TxT matrix $\mathcal{K}_t$ such that

$$\mathbf{X}^{e,t} = \mathcal{K}_t \mathbf{X}$$

where the right-hand side is given by the Kohlhas-Warther law of motion for the forecast as follows:

$$X_s^{e,t} = \bar{f}_t X_{t+(s-t)} = f_{t,s-t}$$

$$= X_{ss} + \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta}\right)^{(t-j)} \frac{1}{1+\delta} dX_s - \gamma \sum_{j=0}^{t} \left(\frac{\delta}{1+\delta}\right)^{(t-j)} \frac{1}{1+\delta} dX_j.$$

It will be convenient to work with a first order perturbation of the Kohlhas-Warther mapping around the steady state – by linearity:

$$d\mathbf{X}^{e,t} = \mathcal{K}_t d\mathbf{X},$$

The final assumption – which holds for the Kohlhas and Warther mapping for $\delta > 0$, is that the expectations on an aggregate input converge to steady state values of the inputs, and hence that the steady-state expectation of an input is the steady-state value of the input. We are now in a position to derive the Jacobian of interest $\hat{J}$ for the Kohlhas-Warther Λ based on the generalised procedure in Bardoczy and Guerreiro (2024) based on the expression for $dY_{i,t}$.

Note that the partial derivatives of the heterogeneous agents mapping evaluated at the steady state with respect to a perturbation to s-th scalar input (including shocks to the true aggregate input sequence, and to time-τ forecasts of period r shocks, for all $\tau, r \leq T$) in the expression for $dY_{i,t}$ corresponds to the following object in Auclert (2021):

$$\frac{\partial Y_{i,t}}{\partial X_s} = \frac{dY_{i,t}^s}{dX_s}$$

in the limit $dX_s \to 0$: that is, to the (t, s) element of the $(T+1) \times (T+2)(T+1)$ heterogeneous block Jacobian $\tilde{J}$, where $t = 0, ..., T$ and $s = 0, ..., (T+2) \times (T+1)$ such that

$$d\mathbf{Y}_i = \tilde{J}[d\mathbf{X}' \quad d\mathbf{X}^{\mathbf{e},\mathbf{0}'} \quad ... \quad d\mathbf{X}^{\mathbf{e},\mathbf{t}'} ... \quad d\mathbf{X}^{\mathbf{e},\mathbf{T}'}]'.$$

Note that by virtue of our assumptions that only the history of expectations up to and including the time t ones, the above Jacobian will have many zero entries: specifically, for each row $t = 0, ..., T$, the entries $[t, s \geq T + T(1+t)]$ will be zeros.

The goal here is to construct a smaller Jacobian $\hat{J}$ with respect to true equilibrium aggregate inputs only (not their expectations) when $dY_{i,t}$ follows the above, based on the FIRE Jacobian J and the Kohlhas-Warther mapping. This target, smaller Jacobian $\hat{J}$ must satisfy

$$d\mathbf{Y}_i = \hat{J} d\mathbf{X}.$$

This implies that

$$\hat{J} d\mathbf{X} = \tilde{J}[d\mathbf{X}' \quad d\mathbf{X}^{\mathbf{e},\mathbf{0}'} \quad ... \quad d\mathbf{X}^{\mathbf{e},\mathbf{t}'} ... \quad d\mathbf{X}^{\mathbf{e},\mathbf{T}'}]'$$

Partition the Jacobian $\tilde{J}$ into $T+2$ sub-Jacobians $\tilde{J}^*, \tilde{J}^0, ..., \tilde{J}^T$ respectively collecting the partial derivatives of the endogenous aggregate output $\mathbf{Y}_i$ with respect to the true input sequence $\mathbf{X}$ and to the each of the forecast sequences at τ, $\mathbf{X}^{e,\tau}$ for $\tau = 0, ..., T$ – i.e. these are the $T+2$ matrices in a block column representation of $\tilde{J}$:

$$\tilde{J} = [\tilde{J}^* \quad \tilde{J}^0 \quad ... \quad \tilde{J}^T].$$

Making use of this block representation, we can rewrite the above condition for $\hat{J}$ as:

$$\hat{J} d\mathbf{X} = \tilde{J}^* d\mathbf{X} + \tilde{J}^0 d\mathbf{X}^{e,0} + ... + \tilde{J}^\tau d\mathbf{X}^{\mathbf{e},\tau} + ... + \tilde{J}^T d\mathbf{X}^{\mathbf{e},\mathbf{T}}$$

And using the linearity of the Kohlhas-Warther mapping for the forecasts/expectations:

$$\hat{J} d\mathbf{X} = \tilde{J}^* d\mathbf{X} + \tilde{J}^0 \mathcal{K}_0 d\mathbf{X} + ... + \tilde{J}^\tau \mathcal{K}_\tau d\mathbf{X} + ... + \tilde{J}^T \mathcal{K}_T d\mathbf{X},$$

$$\hat{J} d\mathbf{X} = \left[\tilde{J}^* + \sum_{\tau=0}^{T} \tilde{J}^\tau \mathcal{K}_\tau\right] d\mathbf{X}$$

Based on this observation, the strategy to recover the Jacobian of interest $\hat{J}$ is as follows: we work on the Jacobian $\tilde{J}$ defined by the expression for $dY_{i,t}$ and consider first the partial derivatives with respect to the true aggregate inputs, then with respect to time t expectations, constructing them as in Auclert (2021) and Bardoczy and Guerriero (2024) based on the FIRE ones, and finally plug these

in together with the Kohlhas-Warther into the above equation. The (t,s) element of $\hat{J}$ will follow by an undetermined coefficients argument from the above equation once we have pinned down the (t,s) element of the matrix $\left[\tilde{J}^* + \sum_{\tau=0}^{T} \tilde{J}^{\tau}\mathcal{K}_{\tau}\right]$. To proceed, consider the following two objects, on which we focus separately: (a) the sub-Jacobian $\tilde{J}^*$ of partial derivatives with respect to the sequence of aggregate inputs, and (b) the finite family sub-Jacobians $\{\tilde{J}^{\tau}\}$ with respect to *forecast* sequence of aggregate inputs formed at each time $\tau \leq T$. A key contribution in Bardóczy & Guerreiro (2024) is how to retrieve these two objects, which we now exploit.

- **The sub-Jacobian $\tilde{J}^*$ of partial derivatives with respect to *true* aggregate inputs**

Throughout, the key observation is that for the model at hand the recursive dynamic programming problem defining, together with the aggregation and distributional law of motion rules, the mapping h only depends on the current true inputs and expectations of the future (one period ahead) inputs. That is, denoting with $\mathcal{V}$ the standard expectational Bellman operator

$$v_t = \mathcal{V}(v_{t+1}^{e,t}, X_t)$$

$$v_s^{e,t} = \mathcal{V}(v_{s+1}^{e,t}, X_{s+1}^{e,t})$$

Now consider the partial equilibrium object $dY_{i,t}^s$, that is the deviation of the endogenous aggregate output from its value along the steady state sequence when the only shocked *true aggregate input* is the time-s one and $dX_r = dX_r^{e,t} = 0, \forall r \neq s$. If $s > t$, by definition of the recursive DP problem in the mapping h we have that

$$v_t = \mathcal{V}(v_{t+1}^{e,t}, X_{ss})$$

$$v_{t+1}^{e,t} = \mathcal{V}(v_{t+2}^{e,t}, X_{ss})$$

Hence, with the standard boundary/terminal value condition $v_T^{e,t} = v_T = v$:

$$v_t = \mathcal{V}(v, X_{ss}),$$

which implies no deviations from the steady state optimal policy. Because an analogous argument holds for any $t' < t$, then there cannot have been adjustment to the time-t beginning of period distribution from optimisation in previous periods, and because the distribution at the origin of the transition is the steady state one (under our assumed boundary conditions), necessarily $dY_{i,t}^s = 0$ for $s > t$. Consider now $dY_{i,t}^s$ when $t \geq s$. As in Auclert et al (2021), a first order perturbation of the aggregation rule/discretised integration $Y_{i,t}^s = {y_{i,t}^s}' D_t^s$ around the steady state gives:

$$dY_{i,t}^s = {dy_{i,t}^s}' D_{ss} + y_{ss}' dD_t^s.$$

Now, if $t = s$ it follows by our analysis of the case $t < s$ that $\forall r < t < s, dy_r^s = d\Lambda_r^s = 0$, implying $dD_t^s = 0$. Hence for $t = s$:

$$dY_{i,t}^s = {dy_{i,t}^s}' D_{ss} = dy_{i,T-1-(s-t)}^{T-1} = \mathcal{Y}_0 dx$$

where the second equality follow by stationarity of the DP problem. As in Auclert, $\mathcal{Y}_0$ is the first element of the FIRE Jacobian and the of the Fake-News matrix. By contrast, if $t > s$ again by definition of the recursive DP problem and of $dY_{i,t}^s$, it follows that $dy_{i,t}^s = 0$ for all $t > s$ since optimal policies have gone back to the steady state, and any impact of the shocked X_s can only arise from propagation through the distribution. For $t \geq s + 1$:

$$dY_{i,t}^s = y_{ss}' dD_t^s$$

where the around the steady state the distribution is governed by the linearised/first order perturbed Chapman-Kolgomorov equation:

$$dD_t^s = \Lambda_{ss}' dD_{t-1}^s + {d\Lambda_{t-1}^s}' D_{ss} = \Lambda_{ss}' dD_{t-1}^s.$$

The second equality follows from the fact that as noted for $t > s$ the optimal policy and transition functions have reverted to steady state ones. Solving the recursion backwards until the initial shock to the distribution is $dD^s_{s+1} = \Lambda^{s\prime}_s D_s + \Lambda_{ss}{}' dD_s = \Lambda^{s\prime}_s D_{ss}$, we find that

$$dD^s_t = (\Lambda_{ss}{}')^{t-(s+1)} dD^s_{s+1}$$

Hence, for $t > s$, defining as in Auclert et al. (2021) the *expectations vector*[17] $\mathcal{E}_r = \Lambda^r_{ss} y_{ss}$

$$dY^s_{i,t} = y'_{ss}(\Lambda_{ss}{}')^{t-(s+1)} dD^s_{s+1} = \mathcal{E}'_{t-(s+1)} dD^s_{s+1}$$

Then note that, again by stationarity of the DP problem, $dD^s_{s+1} = \Lambda^{s\prime}_s D_{ss} = \Lambda^{0\prime}_0 D_{ss} = dD^0_1 = \mathcal{D}_0 dx$, the latter equality using the definition in Auclert et al (2021). Putting the case $t = s$ and $t > s$ together:

$$dY^s_{i,t} = \begin{cases} \mathcal{Y}_0 dx = \mathcal{Y}_{t-s} dx & \text{if} \quad t = s \\ \mathcal{E}'_{t-s-1} \mathcal{D}_0 dx & \text{if} \quad t > s \end{cases}$$

Which corresponds to the first column of the Jacobian (and Fake News) matrix –compare with Auclert's **Proposition 1** after defining the change of variable $\tau = t - s$ for $t \geq s$.[18] That is:

$$\frac{\partial Y_{i,t}}{\partial X_s} = \frac{dY^s_{i,t}}{dX_s} = \mathcal{F}_{t-s,0} = J_{t-s,0} \qquad \text{for} \quad t \geq s$$

and

$$\frac{\partial Y_{i,t}}{\partial X_s} = 0 \quad \text{for} \quad t < s.$$

The intuitive economic interpretation of the above is that because the shock to the time s input is not part of the agents expectations for $t < s$, and the expectations are kept at the steady ones, then there is no effect until it hits the economy at time s. From time s onwards instead, the behaviour of the economy is the same as the one it would follow from time zero onward when the shock hits at time zero. The last two equations, identical to the ones in Bardoczy and Guerreiro (2024), pin down the first sub-Jacobian $\tilde{J}^*$, which is a lower-triangular matrix. We now turn to $\tilde{J}^\tau$.

- **The sub-Jacobian(s) $\tilde{J}^\tau$ of partial derivatives with respect to forecast aggregate inputs formed at time τ**

We now consider the partial equilibrium object of interest $dY^s_{i,t}$ for $s > T + 1$ – i.e., consistent with the original notation in Auclert et al. (2021) extended for the augmented Bardoczy-Guerreiro representation of the mapping h to include time-varying expectations, $dY^s_{i,t}$ denotes change in the time-t endogenous aggregate when the shocked input is $X^{e,\tau}_r$, that is a shock to X_s where the index is given by $s = (T + 1)(1 + \tau) + 1 + r$.[19]

The first case is the one where $t < \tau$. By our assumption ruling out the use of information not part of the agents time-t information set (expectations formed in the future) when agents optimise at any time t, the elements of the the sub-Jacobian $\tilde{J}^\tau$ for rows $t < \tau$, for all τ, follow immediately:

$$\forall \tau, r, t \quad \frac{\partial Y_{i,t}}{\partial X^{e,\tau}_r} = \frac{dY^{(T+1)(1+\tau)+1+r}_{i,t}}{dX_{(T+1)(1+\tau)+1+r}} = 0 \qquad if \quad t < \tau.$$

Now we consider, for any τ, the case $t \geq \tau$, i.e. the remaining last $T + 1 - \tau$ rows of the sub-Jacobians $\tilde{J}^\tau$. The second implication of the above is that, when considering the impact at time t of a shock to the time $\tau \leq t$ expectations on the time r aggregate input, everything remains at the steady state until time τ. Thus, with any change taking place from τ onwards only, the impact of the shock on the time t output can be analysed in the same way as the impact of a shock to the time $\tau = 0$ expectations on the time $r - \tau$ input. In other words, if with a shock to expectations formed τ the first period in

[17] Sometimes denoted the *Observability Matrix*

[18] Since the two coincide by construction (under Lemma 1) $J_{t,s} = \sum_{r=0}^{\min\{s,t\}} \mathcal{F}_{t-r,s-r}$, which implies $J_{t,0} = \mathcal{F}_{t,0}$

[19] So for example, $dX^{e,0}_0 = dX_{T+2}$, $dX^{e,0}_1 = dX_{T+3}$, and $dX^{e,1}_1 = dX_{2(T+1)+2}$.

which effects are observed is τ, the history before τ does not really matter since it is the steady state path, and consequently the shock is the same shock to time zero expectations, with a change of the timing convention for the economy to $t' = t - \tau$.

With the understanding we can just study shocks $dX_r^{e,0}$, we first consider the impact at the time $t \geq \tau = 0$. For $t = \tau$, it follows from the DP problem of the household and the fact $dX_t = 0$ that, for $r \leq \tau = t$

$$\frac{\partial Y_{i,t}}{\partial X_r^{e,\tau}} = \frac{dY_{i,t-\tau}}{dX_{r-\tau}^{e,0}} = \frac{dY_{i,0}}{dX_0^{e,0}} = 0.$$

The intuition is that because the expectations formed at $t = \tau$ on the time t or previous inputs do not matter conditional on observing the time-t input, and the latter is at its steady state value, then shocks to $X_0^{e,0}$ elicit no impact on $Y_{i,0}$, nor by our argument above do shocks $X_\tau^{e,\tau}$ on $Y_{i,\tau}$. By the same logic, it is clear that a shock to the time τ expectation on the time $r \leq \tau$ input $X_\tau^{e,\tau}$ also elicits no impact on $Y_{i,t}$ for $t > \tau$. For any τ, the above equation pins down the entries $(t, r = \tau)$ of $\tilde{J}^\tau$ for rows $t \geq \tau$.

To conclude, within the case $t \geq \tau = 0$, need to consider the impact on Y_t of perturbations to the time $\tau = 0$ expectations on time $r > 0$ inputs, $d\mathbf{X}_{r>0}^{e,0}$.

Consider first the sub-case $t = \tau = 0$. From the DP problem of the household, it is clear that the response is the same as the response to a future shock under FIRE. That is:

$$\forall r > \tau = t, \quad \frac{\partial Y_{i,t}}{\partial X_r^{e,\tau}} = \frac{\partial Y_{i,t-\tau}}{\partial X_{r-\tau}^{e,0}} = \frac{dY_{i,0}^{r-\tau}}{dX_{r-\tau}} = \mathcal{J}_{0,r-\tau} = J_{0,r-t} = \mathcal{Y}_{r-t}$$

Which corresponds to the first row of the FIRE Jacobian and of the Fake-News matrix. Finally, for the case $t > \tau = 0$, note that by construction of the perturbation such that $dX_r = dX_r^{e,\tau'} = 0$ whenever $\tau' \neq \tau$ and of the DP problem, optimal policies have gone back to the steady state, and any impact on the aggregate output must arise from a shock to the beginning of period distribution, i.e. for $t > \tau = 0$ and the only shock being $dX_r^{e,0}$

$$dY_{i,t} = y'_{ss} dD_t^r = y'_{ss}(\Lambda'_{ss})^{t-(\tau+1)} dD_{\tau+1}^r = y'_{ss}(\Lambda'_{ss})^{t-\tau-1} dD_1^{r-\tau}$$

Where the last equality follows again from the fact that prior to time $t = \tau$ the history is the steady state one, so that $dD_{\tau+1}^r = \Lambda_\tau^r D_\tau = \Lambda_\tau^r D_{ss} = \Lambda_{t-\tau}^{r-\tau} D_{ss} = \Lambda_{t-\tau}^{r-\tau} D_{\tau-t} = dD_1^{r-\tau}$. Using the definition of the expectations vector from earlier, recalling $dD_1^{r-\tau} = \mathcal{D}_{r-\tau} dx$:

$$dY_{i,t} = \mathcal{E}_{t-\tau-1} dD_1^{r-\tau} = \mathcal{E}_{t-\tau-1} \mathcal{D}_{r-\tau} dx.$$

Note that this corresponds to the element $F_{t-\tau,r-\tau} dx$ of the (FIRE) Fake-New matrix. Recall from Auclert that such element is defines as:

$$F_{t-\tau,r-\tau} = J_{t-\tau,r-\tau} - J_{t-\tau-1,r-\tau-1}.$$

It follows that entries (t, r) of rows $t > \tau$ of the Jacobian $\tilde{J}^*$ can be constructed from the FIRE Jacobian as:

$$\frac{\partial Y_{i,t}}{\partial X_r^{e,\tau}} = J_{t-\tau,r-\tau} - J_{t-\tau-1,r-\tau-1}$$

- **The Block Jacobian $\tilde{J}$ in terms of the FIRE Jacobian J**

Recall how we partitioned the heterogeneous agents block Jacobian $\tilde{J}$ into column block matrices $\tilde{J} = [\tilde{J}^* \quad \tilde{J}^0 \quad ... \quad \tilde{J}^T]$. We have now pinned down all these column block matrices based on the FIRE Jacobian in Auclert et al. (2021). In particular:

$$\tilde{J}^* = \begin{bmatrix} J_{0,0} & 0 & 0 & ... & 0 \\ J_{1,0} & J_{0,0} & 0 & ... & 0 \\ J_{2,0} & J_{1,0} & J_{0,0} & ... & 0 \\ & & ... & & \\ J_{T,0} & J_{T-1,0} & J_{T-2,0} & ... & J_{0,0} \end{bmatrix}$$

Note that our block matrix $\tilde{J}^*$ – measuring the impact of perturbations to the true inputs on the aggregate output – is equivalent to the *forecast-error* matrix in Bardoczy and Guerreiro (2024). In particular, each column s of their forecast matrix gives the IRF to a forecast error on the time s input that agents learn at time s – here, our formalisation of $\tilde{J}^*$ makes it clear that it is also the IRF to a true shock to the time-s aggregate input when everything else is kept at the steady state. The intuition why the two are (and must be) equivalent is straightforward: because agents do not know or predict their future forecast errors,[20] and the expectations underlying $\tilde{J}^*$ are at their steady states, then a shock to the input dX_s generates a forecast error of the same magnitude that does not hit the economy until time s; when it does, it is the same as an unanticipated time zero shock known at time zero. Turning to the $T+1$ block matrices $\tilde{J}^\tau$, these are given by:

$$\tilde{J}^0 = \begin{bmatrix} 0 & J_{0,1} & J_{0,2} & ... & J_{0,T} \\ 0 & J_{1,1} - J_{0,0} & J_{1,2} - J_{0,1} & ... & J_{1,T} - J_{0,T-1} \\ 0 & J_{2,1} - J_{1,0} & J_{2,2} - J_{1,1} & ... & J_{2,T} - J_{1,T-1} \\ & & ... & & \\ 0 & J_{T,1} - J_{T-1,0} & J_{T,2} - J_{T-1,1} & ... & J_{T,T} - J_{T-1,T-1} \end{bmatrix}$$

$$\tilde{J}^1 = \begin{bmatrix} 0 & 0 & 0 & ... & 0 \\ 0 & 0 & J_{0,1} & ... & J_{0,T-1} \\ 0 & 0 & J_{1,1} - J_{0,0} & ... & J_{1,T-1} - J_{0,T-2} \\ & & ... & & \\ 0 & 0 & J_{T-1,1} - J_{T-2,0} & ... & J_{T-1,T-1} - J_{T-2,T-2} \end{bmatrix}$$

And so forth, so that for an arbitrary $\tilde{J}^\tau$:

1. The first τ rows, i.e. $t < \tau$, are zero row vectors.
2. The first $\tau + 1$ columns, i.e. $s \leq \tau$, are zero column vectors
3. The remaining entries are given by the entries $J_{t,s}$-$J_{t-1,s-1}$ of the FIRE Jacobian, for $0 \leq t \leq T - \tau$ and $0 < s \leq T - \tau$

Note that such entries correspond to those of the FIRE Fake-News matrix: $F_{t,s} = J_{t,s} - J_{t-1,s-1}$ Recalling the equation for the sequence-space partial equilibrium object $dY_{i,t}$, using the above sub-jacobians, we have that:

$$dY_{i,t} = \sum_{s=0}^{t} J_{t-s,0} dX_s + \sum_{\tau=0}^{t-1} \sum_{s=0}^{T} [J_{t-\tau,s-\tau} - J_{t-\tau-1,s-\tau-1}] dX_s^{e,\tau} + \sum_{s=t+1}^{T} J_{0,s-t} dX_s^{e,t}$$

Going a step forward relative to Bardoczy et al (2024), the above can be more easily (and for computational efficiency) be formulated directly in terms of the FIRE Fake-News matrix – without having to actually compute the FIRE Jacobian. To do this, we exploit the fact that, as in Auclert et al (2021):

$$J_{t,s} = \sum_{r=0}^{\min\{s,t\}} F_{t-r,s-r}$$

and our observation that $F_{t,s} = J_{t,s} - J_{t-1,s-1}$. Accordingly:

$$dY_{i,t} = \sum_{s=0}^{t} \mathcal{F}_{t-s,0} dX_s + \sum_{\tau=0}^{t-1} \sum_{s=0}^{T} \mathcal{F}_{t-\tau,s-\tau} dX_s^{e,\tau} + \sum_{s=t+1}^{T} \mathcal{F}_{0,s-t} dX_s^{e,t}$$

[20] As we assumed, a law of iterated expectations holds to the effect that projections on the information set in any given period τ – i.e. τ time expectations under internal rationality – are orthogonal to every element in the information set. If the evolution of information sets obeys a natural filtration, then they are also orthogonal to the information set at all time $s \leq \tau$. As a consequence, future forecast error cannot be predicted and they must hit as unanticipated time zero shocks

$$dY_{i,t} = \sum_{s=0}^{t} \mathcal{F}_{t-s,0} dX_s + \sum_{\tau=0}^{t-1} \sum_{s=0}^{T} \mathcal{F}_{t-\tau,s-\tau} \mathcal{K}_{\tau,s} d\mathbf{X} + \sum_{s=t+1}^{T} \mathcal{F}_{0,s-t} \mathcal{K}_{t,s} d\mathbf{X}$$

Where $\mathcal{F}_{t-s,0}$ correspond to entries of the *first column* of the Fake-News matrix with row index $t-s$, and $F_{0,t-s}$ correspond to the entries of the same matrix for the zeroth row and with column index $t-s$. $\mathcal{K}_{\tau,s}$ denotes row s of the Kohlhas-Warther time-varying mapping $\mathcal{K}_\tau$ at time τ. From the above equation, it follows that the $(t,r)-th$ element of the Jacobian $\hat{J}$ of the heterogeneous agent block output with respect an arbitrary true equilibrium input dX_s is:

$$\hat{J}_{t,r} = \frac{\partial Y_{i,t}}{\partial X_s} = \frac{dY_{i,t}}{dX_s} = \mathcal{F}_{t-r,0} + \sum_{\tau=0}^{t-1} \sum_{s=0}^{T} \mathcal{F}_{t-\tau,s-\tau} \mathcal{K}_{\tau,s,r} + \sum_{s=t+1}^{T} \mathcal{F}_{0,s-t} \mathcal{K}_{t,s,r}$$

Where $\mathcal{K}_{\tau,s,r}$ denotes the (s,r)-th element of the matrix $\mathcal{K}_\tau$, and given generally (see earlier section) by:

$$\mathcal{K}_{\tau,s,r} = \frac{\partial f_\tau X_s}{\partial x_r} = \begin{cases} \sum_{j=0}^{\tau} \left(\frac{\delta}{1+\delta}\right)^{\tau-j} \frac{1}{1+\delta} & if \quad s > \tau, r = s \\ -\gamma \left(\frac{\delta}{1+\delta}\right)^{\tau-r} \frac{1}{1+\delta} & if \quad s > \tau, r \leq \tau \\ 1 & if \quad s \leq \tau, r = s \\ 0 & \text{otherwise} \end{cases}$$

This approach to constructing the Jacobian $\hat{J}$, only using the elements of the Fake-News and the parameters of the Kohlhas-Warther mapping, is equivalent to the generalised procedure in Bardoczy and Guerreiro defined by the mapping in section 2.1.2 but only efficiently requires knowledge of the FIRE Fake-News matrix, a computationally primitive object with respect to the FIRE Jacobian since the latter is defined by recursive sums over the relevant diagonals of the former.

A.4 Generalisation to Time-Varying Coefficients γ_k, δ_k

The Kohlhas-Warther reduced form equation for the evolution of the forecast with time-varying extrapolation and news-revision coefficients is:

$$x_{t+k} - f_{it}x_{t+k} = \alpha_{i,k} + \gamma_k x_t + \delta_k(\bar{f}_t x_{t+k} - \bar{f}_{t-1}x_{t+k}) + \varepsilon_{i,t|t+k}.$$

Relaxing the framework presented in the main text, the above equation allows for the strength of the extrapolation and news-revision effects to vary in function of the time elapsed between the time at which the forecast is formed and the timing of the forecast variable. This also implies that the strength of extrapolation and news-revision is the same at times t and r when respectively forecasting x_{t+k} and x_{r+k}. Again, let $f_t x_{t+k} := \int_j f_{jt}x_{t+k}dP(j)$ denote the average expectation in the cross-section. Integrating over the measure of households and re-arranging:

$$x_{t+k} - \bar{f}_t x_{t+k} = \alpha_{i,k} + \gamma_k x_t + \delta_k(\bar{f}_t x_{t+k} - \bar{f}_{t-1}x_{t+k}).$$

$$\bar{f}_t x_{t+k} = c_k + \frac{1}{1+\delta_k}(\delta_k \bar{f}_{t-1}x_{t+k} + x_{t+k} - \gamma_k x_t),$$

where $c_k = -\frac{1}{1+\delta_k}\int_j \alpha_{j,k} + \epsilon_{j,t|t+k}P(dj) = -\frac{1}{1+\delta_k}\int_j \alpha_{j,k}P(dj)$. We impose again a no-bias assumption on the constant term c_k for all $k \in N$, so that $\bar{f}_t x_{ss} = \bar{f}_{t-1}x_{ss} = \bar{f}_{ss}x_{ss} = x_{ss}$:

$$c_k = \frac{\gamma_k}{1+\delta_k}x_{ss}.$$

Note that, using the definition of c_k the no-bias assumption implies a restriction on the intercept or household fixed effects, $\forall k$:

$$\gamma_k x_{ss} + \int_j \alpha_{j,k}P(dj) = 0.$$

One interpretation of the above is that the bias of the agents when forming individual forecasts exactly offsets the extrapolation effect at the steady state. The reason is that, as the news revision channel ceases to play a role at the steady state (the public information set becomes stationary), then an unbiased aggregate forecast requires that agents adjust their idiosyncratic bias so as to offset aggregate extrapolation. For an aggregate input with a positive-valued steady state, if we require $\gamma_k \leq 0$, consistent with the micro-founded model of extrapolation from asymmetric attention and empirical estimates in Kohlas and Warther (2021), then the no-bias assumption implies

$$\int_j \alpha_{j,k}P(dj) < 0.$$

With this in mind, the generalisation of the Kohlhas-Warther mapping to time-varying extrapolation and news revision parameters proceeds in the same way as for the case with time-invariant ones. At a first order perturbation around the non-stochastic steady state, the forecast at time t of an aggregate input k periods ahead satisfies:

$$\bar{f}_t(x_{ss} + dx_{t+k}) = c_k + \frac{1}{1+\delta_k}\left(\delta_k \bar{f}_{t-1}(x_{ss} + dx_{t+k}) + x_{ss} + dx_{t+k} - \gamma_k(x_{ss} + dx_t)\right)$$

Using the no-bias assumption pinning down each c_k:

$$\bar{f}_t(x_{ss} + dx_{t+k}) = x_{ss} + \frac{1}{1+\delta_k}\left(\delta_k \bar{f}_{t-1}dx_{t+k} + dx_{t+k} - \gamma_k dx_t\right)$$

$$\bar{f}_t dx_{t+k} = \frac{1}{1+\delta_k}\left(\delta_k \bar{f}_{t-1}dx_{t+k} + dx_{t+k} - \gamma_k dx_t\right)$$

This yields, like previously, a first order difference equation in the forecasts formed at times $t = 0...T$. Solving the equation backwards, together with the usual boundary conditions:

$$\bar{f}_t dx_{t+k} = \frac{1}{1+\delta_k}\left[\delta_k \frac{1}{1+\delta_{k-1}}\left(\delta_{k-1}\bar{f}_{t-1}dx_{t+k} + dx_{t+k} - \gamma_{k+1}dx_{t-1}\right) + dx_{t+k} - \gamma_k dx_t\right]$$

$$\bar{f}_t dx_{t+k} = \sum_{j=0}^{t}\left(\prod_{s=j}^{t-1}\frac{\delta_{k+1-(t-s)}}{1+\delta_{k-(t-s)}}\right)\frac{1}{1+\delta_k}dx_{t+k} - \sum_{j=0}^{t}\gamma_{k+(t-j)}\left(\prod_{s=j}^{t-1}\frac{\delta_{k+1-(t-s)}}{1+\delta_{k-(t-s)}}\right)\frac{1}{1+\delta_k}dx_j.$$

B Two Asset HANK Model

B.1 Model Equations

This appendix collects the equilibrium conditions of the model.

B.2 Households

There are six groups of households $g \in \{1, \ldots, 6\}$ with population shares ω_g and group-specific discount factors β_g. Within each group, households face idiosyncratic productivity risk z governed by a Markov chain with transition matrix Π. Households choose consumption c and liquid asset holdings a' subject to a borrowing constraint $a' \geq \underline{a}$.

Budget constraint. For a household of type g with productivity z and liquid assets a:

$$c + a' = (1 + r^p)a + (1 - \tau)w\, h\, e_g\, z + d_g + T^{gov} \tag{B.1}$$

where r^p is the return on liquid assets, τ is the labour income tax rate, w is the real wage, h is hours worked (determined by the union, see below), e_g is the group-specific income share, d_g is the distribution from the illiquid portfolio to group g, and T^{gov} is a lump-sum government transfer.

Euler equation (liquid asset). The household's optimality condition is:

$$c^{-1/\sigma} \geq \tilde{\beta}_g\, E_t \left[(1 + r^p_{t+1})\, c^{-1/\sigma}_{t+1} \right], \quad \text{with equality if } a' > \underline{a} \tag{B.2}$$

where σ is the elasticity of intertemporal substitution and $\tilde{\beta}_g = \beta_g \cdot e^{\eta_{C,t}}$ incorporates the discount factor shock.

Aggregation. Aggregate consumption C_t, liquid asset demand A_t, disposable income Y^S_t, and the marginal rates of substitution MRSC_t, MRSH_t are obtained by integrating over the stationary distribution:

$$C_t = \sum_g \frac{\omega_g}{\mu_g} \int c_g(z, a)\, \mathrm{d}\mu_g(z, a), \qquad A_t = \sum_g \frac{\omega_g}{\mu_g} \int a'_g(z, a)\, \mathrm{d}\mu_g(z, a) \tag{B.3}$$

where μ_g is the within-group measure.

B.2.1 Illiquid Asset and Portfolio Inattention

Illiquid asset law of motion.

$$X_t = (1 + r^{ill}_t)X_{t-1} - d_t \tag{B.4}$$

where X_t is aggregate illiquid wealth, r^{ill}_t is the return on illiquid assets, and d_t is the aggregate distribution from illiquid to liquid accounts.

Distribution policy.

$$d_t = \frac{\bar{r}^{ill}}{1 + \bar{r}^{ill}}\, \tilde{X}_t + \chi \left(\tilde{X}_t - (1 + \bar{r}^{ill})\bar{X} \right) \tag{B.5}$$

where $\tilde{X}_t$ denotes expected illiquid wealth (defined below) and χ governs the sensitivity of distributions to perceived deviations from steady state.

Sticky portfolio expectations. Portfolio expectations update infrequently with probability $(1-\theta)$ per period:

$$\tilde{X}_t = (1 - \theta)(1 + r^{ill}_t)X_{t-1} + \theta\, \tilde{X}_{t-1} \tag{B.6}$$

When $\theta = 0$, households have full-information rational expectations over their portfolio: $\tilde{X}_t = (1 + r^{ill}_t)X_{t-1}$.

B.2.2 Production

Final good firm (production function). Output is produced with a Cobb–Douglas technology:

$$Y_t = Z_t\, K_{t-1}^{1-\alpha_h}\, (N_E \cdot h_t)^{\alpha_h} \tag{B.7}$$

where $Z_t = e^{\eta_{z,t}}$ is total factor productivity, K_{t-1} is the capital stock, N_E is aggregate labour efficiency units, and h_t is hours worked. Rearranging for hours:

$$h_t = \frac{1}{N_E} \left(\frac{Y_t}{Z_t\, K_{t-1}^{1-\alpha_h}} \right)^{1/\alpha_h} \tag{B.8}$$

Marginal cost.

$$mc_t = \frac{w_t}{\alpha_h\, Y_t/(h_t\, N_E)} \tag{B.9}$$

Rental rate of capital.

$$r_t^k = mc_t\, (1-\alpha_h)\, \frac{Y_t}{K_{t-1}} \tag{B.10}$$

Dividends of the intermediate firm.

$$\mathrm{Div}_t = (1 - mc_t)\, Y_t \tag{B.11}$$

B.2.3 Capital Firm

Capital accumulation.

$$K_t = I_{t-1} + (1-\delta) K_{t-1} \tag{B.12}$$

Investment Euler equation (Tobin's Q). With convex adjustment costs parameterised by ψ:

$$Q_t = 1 + \frac{\psi}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 + \psi \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} - \frac{\psi}{1 + r_{t+1} + \eta_{rp,t+1}} \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \tag{B.13}$$

Asset price of capital.

$$Q_t = \frac{1}{1 + r_{t+1} + \eta_{rp,t+1}} \left[r_{t+2}^k + (1-\delta) Q_{t+1} \right] \tag{B.14}$$

Capital firm dividends.

$$\mathrm{Div}_t^K = r_t^k K_{t-1} - I_{t-1} - \frac{\psi}{2} \left(\frac{I_{t-1}}{I_{t-2}} - 1 \right)^2 I_{t-1} \tag{B.15}$$

Measured investment.

$$\mathrm{Inv}_t = I_{t-1} \tag{B.16}$$

B.2.4 Nominal Block

Price Phillips curve. With Calvo parameter ζ_p, Kimball super-elasticity ν_p, and demand elasticity ϵ^p:

$$\pi_t = \pi_{t-1} + \frac{(1-\zeta_p)\left(1 - \frac{\zeta_p}{1+\bar{r}}\right)}{\zeta_p} \cdot \frac{\epsilon^p}{\epsilon^p + \nu_p - 1} \left(mc_t - \frac{\epsilon^p - 1}{\epsilon^p} \right) + \frac{1}{1 + r_t} (\pi_{t+1} - \pi_t) + \eta_{p,t} \tag{B.17}$$

Wage Phillips curve. With Calvo parameter ζ_w, labour supply elasticity (Frisch) φ, and union markup ϵ:

$$\pi_t^w = \pi_{t-1} + \frac{(1-\zeta_w)(1-\bar{\beta}\zeta_w)}{\zeta_w} \cdot \frac{\epsilon}{\epsilon + \nu_p - 1} \left(\phi_l \frac{\text{MRSH}_t}{\text{MRSC}_t} - \frac{\epsilon - 1}{\epsilon} \right) + \bar{\beta} \left(\pi_{t+1}^w - \pi_t \right) + \eta_{w,t} \tag{B.18}$$

where ϕ_l is a steady-state labour disutility scaling parameter and $\bar{\beta}$ is the average discount factor.

Real wage evolution.

$$\ln\left(w_t(1+\pi_t)\right) - \ln w_{t-1} = \pi_t^w \tag{B.19}$$

Taylor rule.

$$i_t = \rho_m \, i_{t-1} + (1-\rho_m) \left[r^* + \phi_\pi \, \pi_t + \phi_y \frac{Y_t - \bar{Y}}{\bar{Y}} \right] + \eta_{r,t} \tag{B.20}$$

Fisher equation.

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}} \tag{B.21}$$

B.2.5 Asset Pricing and Returns

Return on liquid assets.

$$r_t^p = r_{t-1} - \xi \tag{B.22}$$

where ξ is the financial intermediation spread.

Government bond price. With maturity structure governed by decay parameter δ_b:

$$q_t^b = \frac{1 + \delta_b \, q_{t+1}^b}{1 + r_t} \tag{B.23}$$

Equity prices. No-arbitrage conditions for the intermediate firm and capital firm:

$$p_t = \frac{p_{t+1} + \text{Div}_{t+1}}{1 + r_t + \eta_{rp,t}} \tag{B.24}$$

$$p_t^K = \frac{p_{t+1}^K + \text{Div}_{t+1}^K}{1 + r_t + \eta_{rp,t}} \tag{B.25}$$

Fiscal Policy

Government spending.

$$G_t = \bar{G} + \bar{Y} \, \eta_{g,t} \tag{B.26}$$

Tax rule.

$$\tau_t = \bar{\tau} + \gamma_\tau \, \bar{q}^b \, \frac{B_{t-1} - \bar{B}}{\bar{Y}} \tag{B.27}$$

Government budget constraint.

$$q_t^b B_t = G_t + (1 + \delta_b q_t^b) B_{t-1} + T^{gov} - \tau_t \left(Y_t - r_t^k K_{t-1} - \text{Div}_t \right) \tag{B.28}$$

B.2.6 Market Clearing

Asset market.

$$A_t + X_t = q_t^b B_t + p_t + p_t^K \tag{B.29}$$

Goods market.

$$Y_t = C_t + G_t + I_{t-1} + \frac{\psi}{2}\left(\frac{I_{t-1}}{I_{t-2}} - 1\right)^2 I_{t-1} + \xi\, A_{t-1} \tag{B.30}$$

Funds market.

$$(1 + r_t^p + \xi)A_{t-1} + (1 + r_t^{ill})X_{t-1} = p_t + \mathrm{Div}_t + p_t^K + \mathrm{Div}_t^K + (1 + \delta_b q_t^b)B_{t-1} \tag{B.31}$$

B.2.7 Exogenous Shock Processes

Five of the seven structural shocks follow AR(1) processes:

$$\eta_{C,t} = \rho_C\, \eta_{C,t-1} + \varepsilon_t^C \qquad \text{(discount factor)} \tag{B.32}$$
$$\eta_{rp,t} = \rho_{rp}\, \eta_{rp,t-1} + \varepsilon_t^{rp} \qquad \text{(risk premium)} \tag{B.33}$$
$$\eta_{z,t} = \rho_z\, \eta_{z,t-1} + \varepsilon_t^z \qquad \text{(TFP)} \tag{B.34}$$
$$\eta_{r,t} = \rho_r\, \eta_{r,t-1} + \varepsilon_t^r \qquad \text{(monetary policy)} \tag{B.35}$$
$$\eta_{g,t} = \rho_g\, \eta_{g,t-1} + \varepsilon_t^g \qquad \text{(government spending)} \tag{B.36}$$

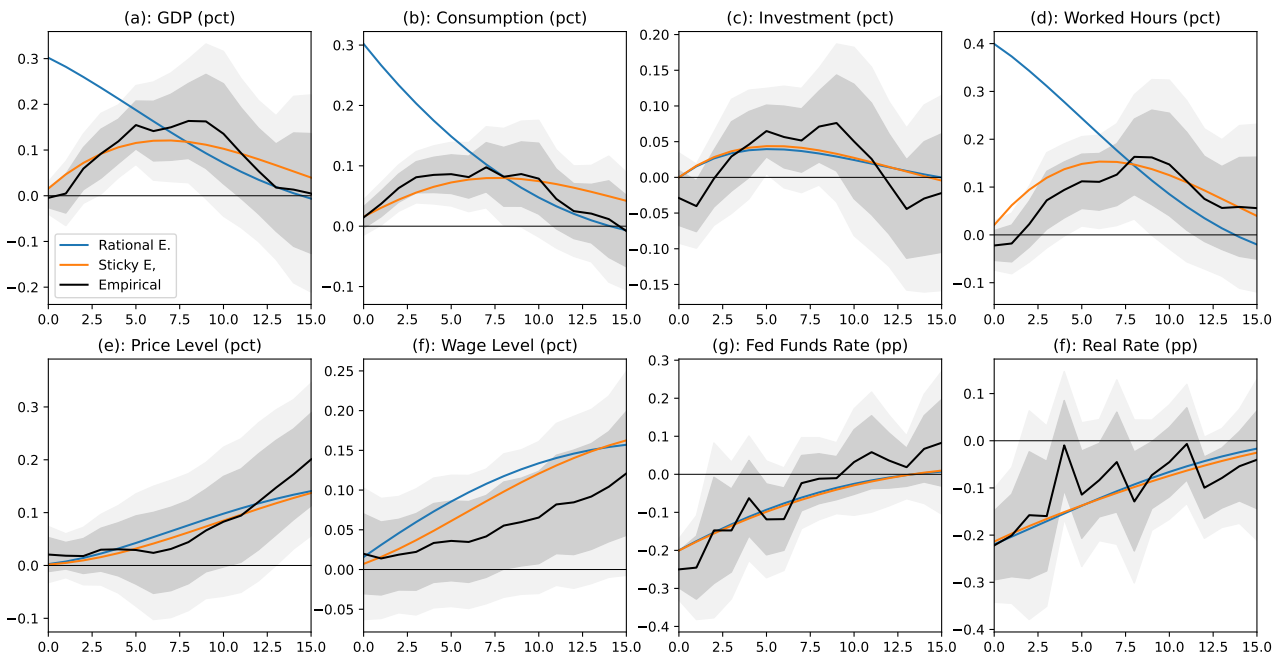
The price and wage markup shocks follow ARMA(1,1) processes as in Smets & Wouters (2007):

$$\eta_{p,t} = \rho_p\, \eta_{p,t-1} + \varepsilon_t^p - \theta_p\, \varepsilon_{t-1}^p \qquad \text{(price markup)} \tag{B.37}$$
$$\eta_{w,t} = \rho_w\, \eta_{w,t-1} + \varepsilon_t^w - \theta_w\, \varepsilon_{t-1}^w \qquad \text{(wage markup)} \tag{B.38}$$

All innovations $\varepsilon_t^j \sim \mathcal{N}(0, \sigma_j^2)$ are mutually independent.

B.3 Model Dynamics

Figure B.1: Replicating Auclert et al. (2020)



Note: This figure replicates the impulse response to a monetary policy shock under rational and stictky expectations as in Auclert et al. (2020). The sticky impulse responses were IRF matched to the empirical evidence of a local projection to a Romer & Romer (2004) monetary policy shock.

B.4 Business Cycle Estimation

B.4.1 Data and Estimation

This appendix describes the construction of the dataset used to estimate the model. The sample runs from 1978 Q2 to 2018 Q4. The first seven observables follow the transformations of Smets & Wouters (2007) closely; two additional series capture survey expectations of inflation for household and professionals.

All macroeconomic aggregates are quarterly and sourced from FRED. The raw series are: nominal GDP, nominal personal consumption expenditures, nominal business fixed investment, compensation per hour (nonfarm business sector), an index of aggregate hours (average weekly hours × civilian employment), the GDP deflator, civilian non-institutional population, and the effective federal funds rate. Real quantities are deflated by the GDP deflator throughout.

Survey expectations are drawn from two sources:

- **Survey of Professional Forecasters (SPF):** individual-level microdata from the Philadelphia Fed. We use the four-quarter-ahead forecasts of the GDP deflator, computed as $100 \times (X_{t+4}/X_t - 1)$ from each respondent's reported level forecasts, then averaged across forecasters in each quarter.
- **Michigan Survey of Consumers:** individual-level microdata on expected price changes over the next 12 months (`px1`), aggregated to a quarterly frequency using the survey sampling weights.

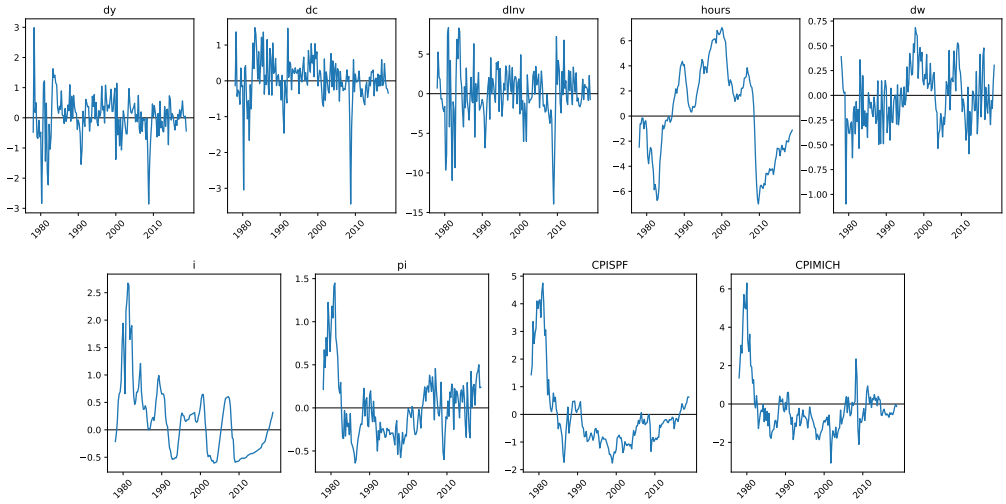
Table B.1 summarises the nine observables. All series are linearly detrended (OLS residuals from a regression on a linear time trend) rather than simply demeaned; this accounts for low-frequency movements in trend growth and the secular decline in nominal interest rates over the sample.

Variable	**Transformation**
Δy_t	$100 \times \left[\ln(GDP_t/P_t/N_t) - \ln(GDP_{t-1}/P_{t-1}/N_{t-1})\right]$, linearly detrended
Δc_t	$100 \times \left[\ln(C_t/P_t/N_t) - \ln(C_{t-1}/P_{t-1}/N_{t-1})\right]$, linearly detrended
Δi_t	$100 \times \left[\ln(I_t/P_t/N_t) - \ln(I_{t-1}/P_{t-1}/N_{t-1})\right]$, linearly detrended
h_t	$100 \times \ln(\bar{h}_t \cdot E_t/N_t)$, linearly detrended
Δw_t	$100 \times \left[\ln(W_t/P_t) - \ln(W_{t-1}/P_{t-1})\right]$, linearly detrended
i_t	Federal funds rate divided by 4 (quarterly), linearly detrended
π_t	$100 \times \ln(P_t/P_{t-1})$, linearly detrended
CPISPF_t	SPF mean four-quarter-ahead GDP deflator inflation forecast, linearly detrended
CPIMICH_t	Michigan Survey weighted-mean expected price change (next 12 months), linearly detrended

Table B.1: Observable variables and their transformations. P_t denotes the GDP deflator, N_t is civilian non-institutional population (normalised to an index), E_t is civilian employment (normalised), $\bar{h}_t$ is average weekly hours (normalised), and W_t is nominal compensation per hour.

B.4.2 Further Results

Figure B.2: Business cycle data



Note: Figure plots the transformed data used to estimate the Two Asset Hank Model. The sample range spans 1978 to 2018.

Figure B.3: Parameter chains

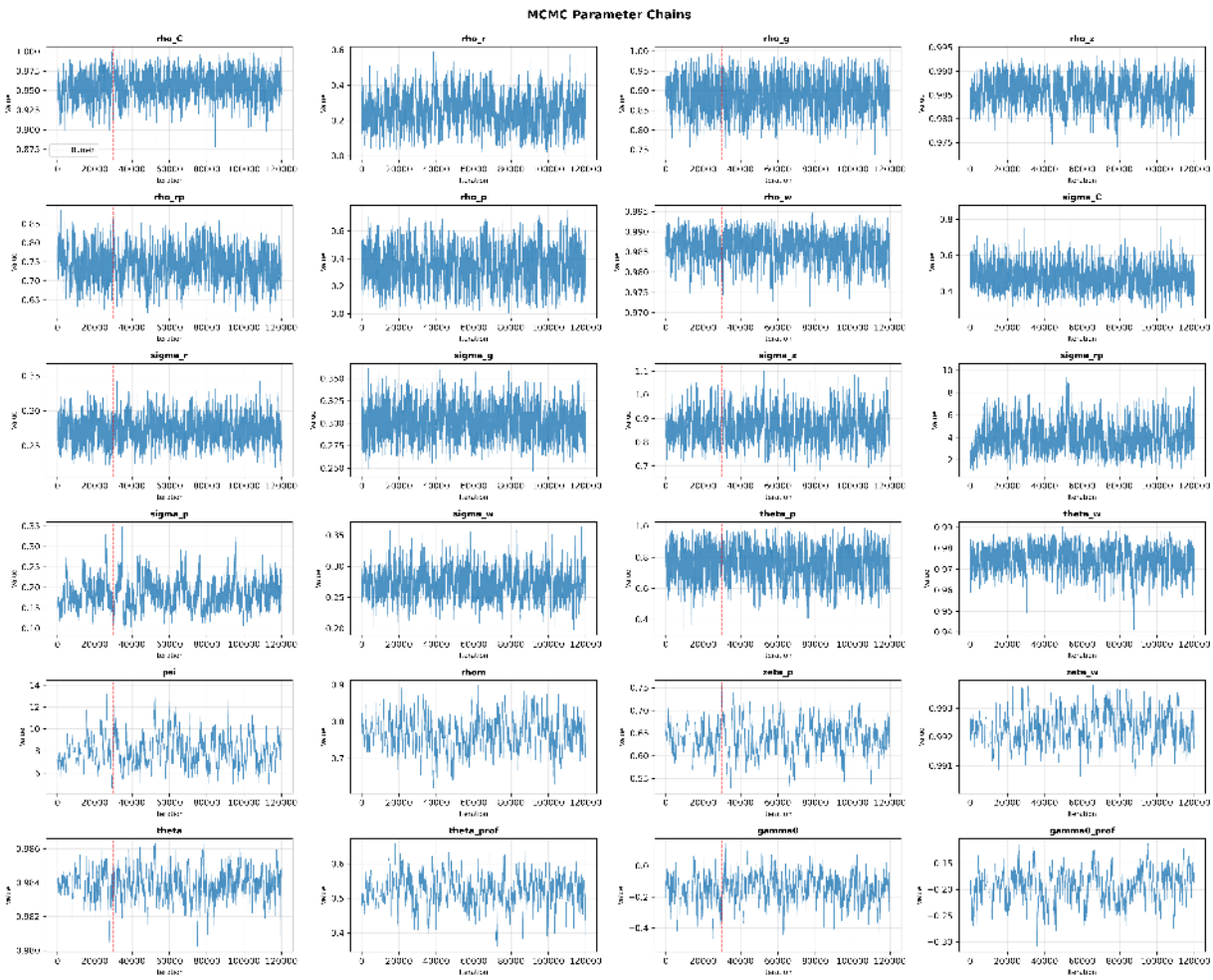


Figure B.4: Geweke convergence test

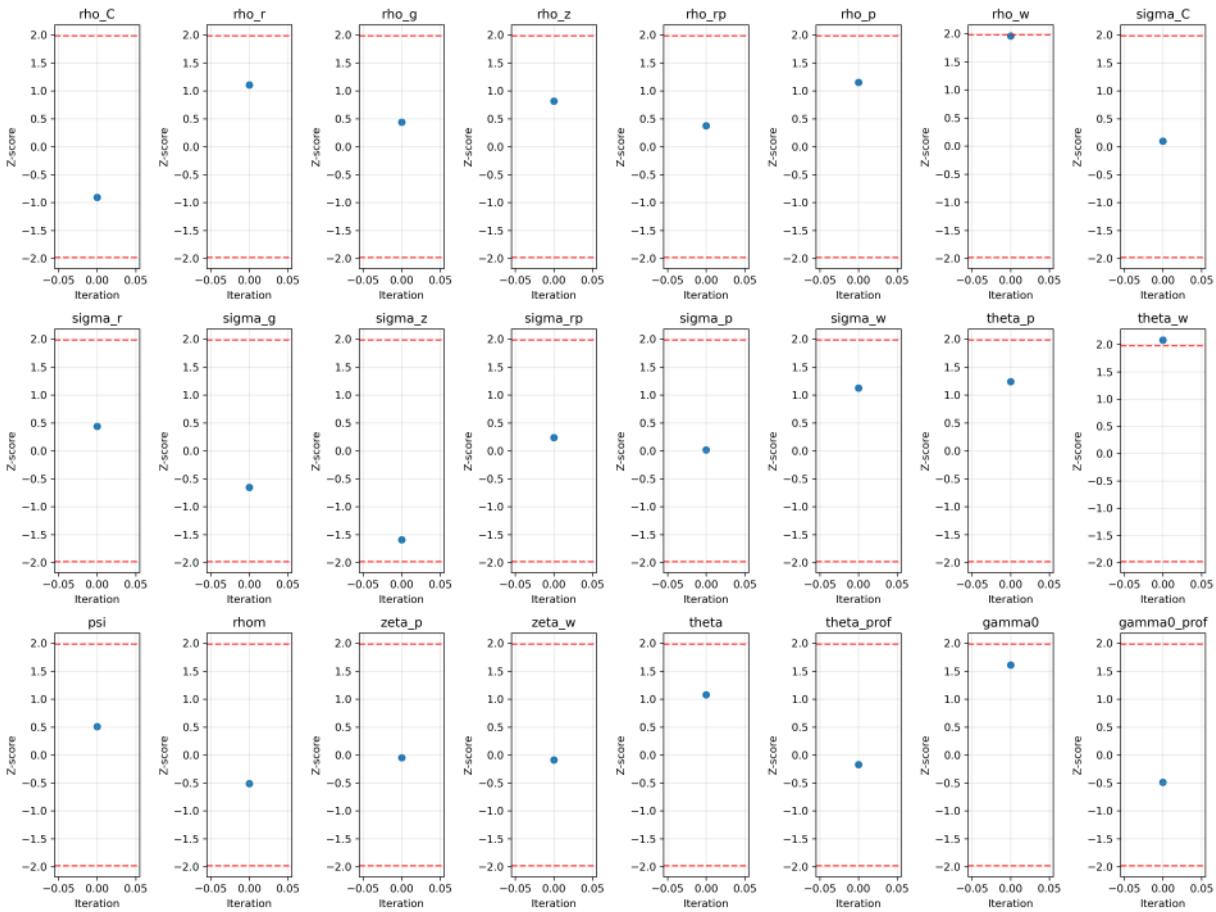
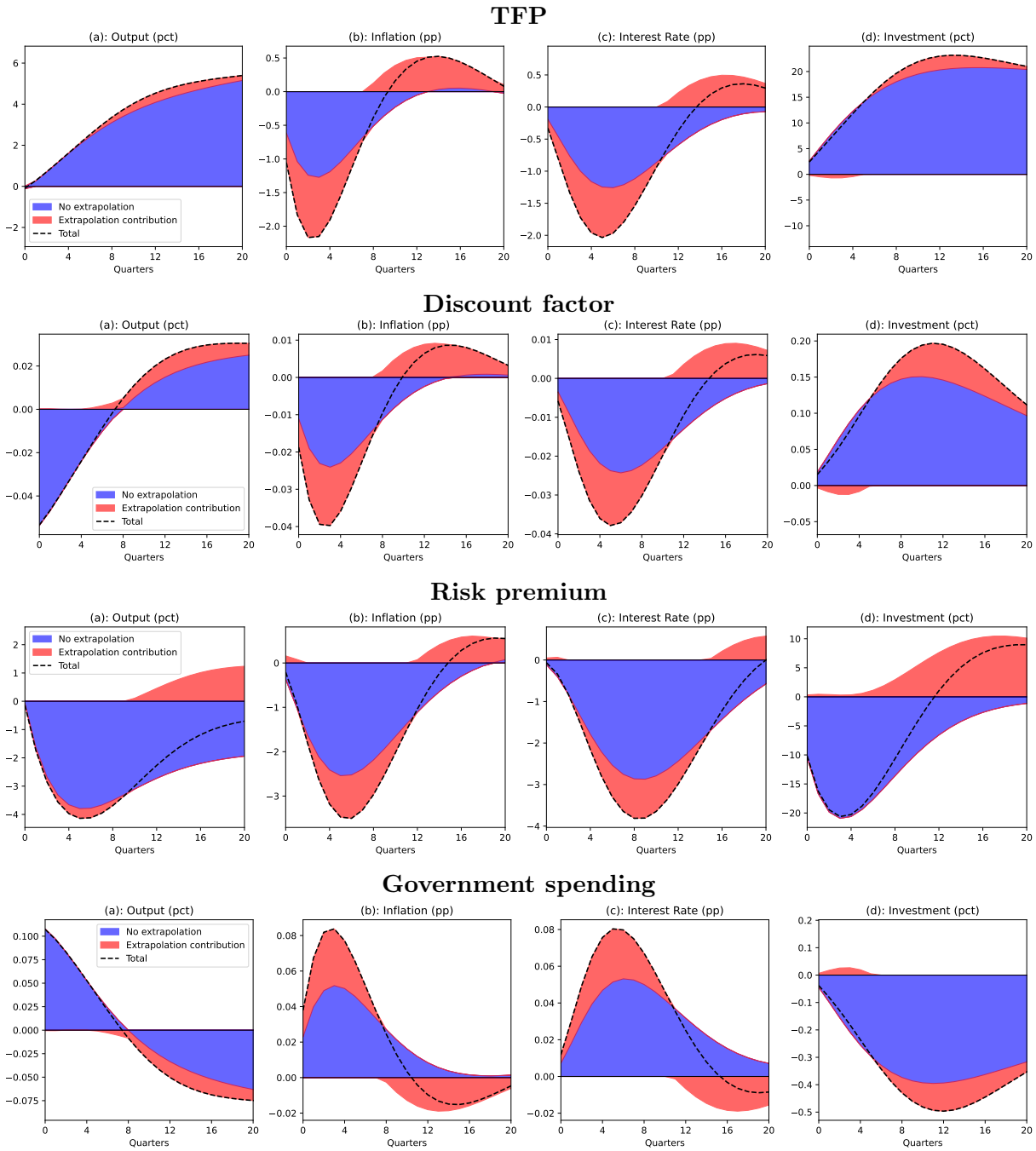
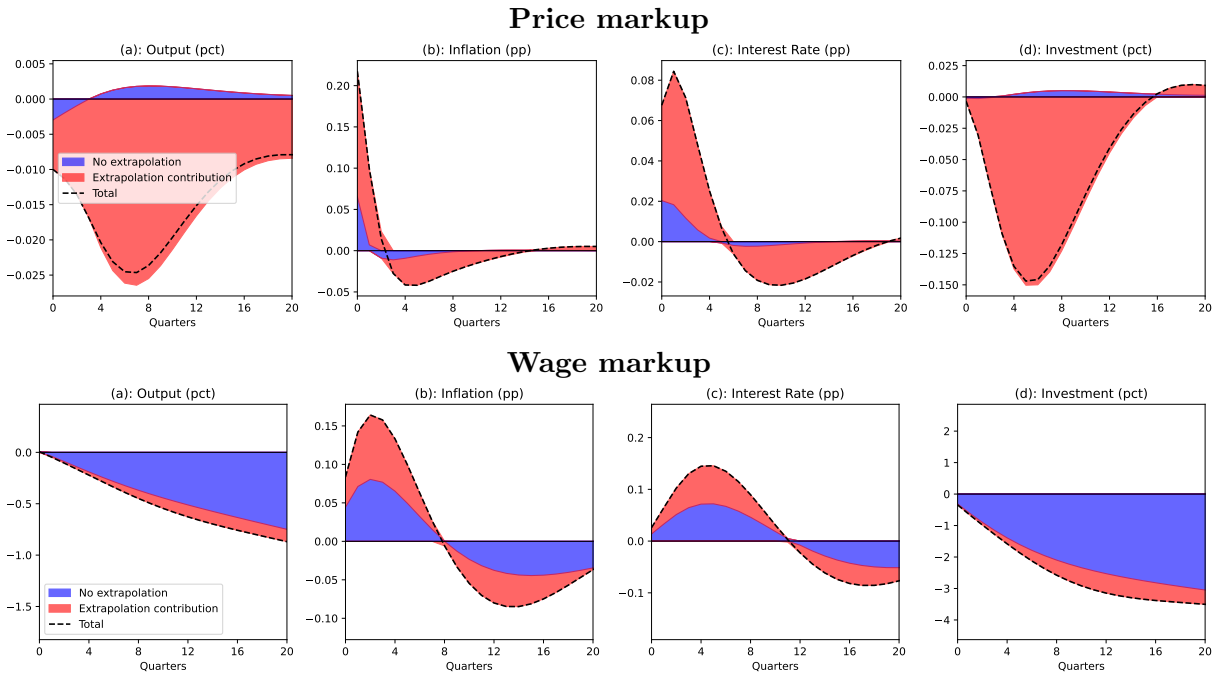


Figure B.5: Other Shocks





Note: Impulse response to a one standard deviation shock decomposed as described in Figure 1.

C Model-implied forecast-errors with approximate $MA(\infty)$ representations

As part of the proposed applications, we are interested in evaluating the properties of the model-implied forecast under behavioural expectations versus the nested purely sticky expectations case, particularly the model-implied forecast-error and its variance, both theoretical and against business-cycle data. To back out the forecast error, we rely on the approximate $MA(\infty)$ representation in the innovations of the K aggregate shocks, which is closely linked to the general equilibrium SSJ $\tilde{\mathbf{G}}$. In particular, the coefficients in the MA representation in the (Gaussian) innovations vector $\mathbf{e}^Z = [\epsilon^{Z^1} \quad \epsilon^{Z^2} \quad ...\epsilon^{Z^K}]'$ of any endogenous output can be obtained from the simulation of its IRFs to a pulse in the innovations at the origin Auclert et al. (2021), on the assumption a suitable MA representation exists for each of the K aggregate shock process $d\mathbf{Z}^k$ of the form:

$$dZ_t^k = \sum_s M_s^{Z^k} \epsilon_{t-s}^{Z^k}$$

where $[M_s]_{s=0...}$ are the MA coefficients. In particular, letting $G^{X,Z}$ mapping the sequence in the aggregate shock $Z \in \mathcal{Z}$ to the sequence on the arbitrary endogenous output $d\mathbf{X}$, the $MA(T-1) \to MA(\infty)$ representation of the endogenous output at time t is:

$$dX_t = \sum_{Z \in \mathcal{Z}} G_{t,j\geq 0}^{X,Z} d\mathbf{Z} = \sum_{Z \in \mathcal{Z}} G_{t,j\geq 0}^{X,Z} \left[\sum_s M_s^Z \epsilon_{j-s}^Z \right]'_{j=0...T} = \sum_{Z \in \mathcal{Z}} \sum_j \sum_s G_{t,j}^{X,Z} M_s^Z \epsilon_{j-s}^Z.$$

For some arbitrary time t and lag $k \geq 0$, consider the impact of an innovation at time ϵ_{t-k}, orthogonal to the full information set at times prior to $t-k$, on the time t output dX_t – that is the MA coefficient M_k^X. This is the same as the impact of an innovation in period time ϵ_0 on the output dX_k.[21] Hence for any lag k, the MA coefficient M_k^X on ϵ_{t-k}^Z can be found as the coefficient on $\epsilon_0^Z = \epsilon_{j-s}^Z$ for $s = j$ in the MA representation of dX_k:

$$M_k^X = \sum_j G_{k,j}^{X,Z} M_j^Z = G_{k,j\geq 0}^{X,Z} M^Z,$$

[21] Since the shock is orthogonal to all information sets dated strictly lower than $\tau = t-k$ and hence unanticipated before period $\tau < t$, its impact on output is equivalent to that of a time-zero shock on outputs $dX_{t-(t-k)} = dX_k$.

Where the last equality uses the fact we can collect the MA coefficients for the aggregate shock vector $d\mathbf{Z}$ into a column vector $M^Z = [M_0 \quad M_1 \quad ... \quad M_{T-1}]$ corresponding, by construction of the MA representation, to the IRF of the dZ process to a pulse in the innovation. Stacking over the M_k^X coefficients into a row vector yielding the MA representation of dX_t for all t by making use of the definition of the SSJ Jacobian $G^{X,Z}$:

$$M^{X,Z} = G^{X,Z} M^Z$$

The above makes it clear why computationally the MA representation can be obtained in two equivalent ways:

1. A simulated IRF against obtained by mapping a simulated sequences $d\mathbf{Z}$, for $Z \in \mathcal{Z}$, obtained through pulses in the respective time-zero innovations ϵ_0^Z
2. When the aggregate shocks process is included in the model blocks and model specified in the innovations, as the first column of a Jacobian since under the assumption of i.i.d. Gaussian innovations $M^Z = [1 \quad 0 \quad ... \quad 0]'$.[22]

Theoretical model-implied forecast error and its variance

For an arbitrary endogenous output process and time t, we define as the *j-steps ahead forecast error* the residual of the time $t+j$ output on its projection on the time-t information set of the econometrician with knowledge of both the model and the history of innovations ϵ^Z through time t. That is, analogously to the j-step ahead innovations against prior expectations (or projections) used in the Kalman-Filter recursion (Harvey, 2014; Anderson and Moore, 1979):

$$v_t^j = dX_{t+j} - dX_{t+j|t}$$
$$\text{where} \quad dX_{t+j|t} = E_t dX_{t+j} = \sum_{Z \in \mathcal{Z}} \sum_k M_k^{X,Z} E_t \epsilon_{t+j-k}^Z.$$

The second equality follows from the definition of the MA representation for the process dX at time $t+j$ and the *projection theorem*, defining the efficient projection in a minimum variance sense. The forecast function above is by construction measurable with respect to the information set of the econometrician at time t and hence without look-ahead bias. By construction of the information set and the law of iterated expectations, $E_t \epsilon_{t+j-k}^Z = 0$ for $j-k > 0$ while $E_t \epsilon_{t+j-k}^Z = \epsilon_{t+j-k}$ for $j-k \leq 0$. Combining these observations, it follows that the j-steps ahead forecast error depends intuitively only on the reaction to shocks arriving strictly between t and $t+j$, which propagate at time $t+j$ as a shock at time zero on the output at time j, and that the history of shocks is irrelevant:

$$v_t^j = \sum_{Z \in \mathcal{Z}} \sum_k M_k^{X,Z} \epsilon_{t+j-k}^Z - \sum_{Z \in \mathcal{Z}} \sum_k M_k^{X,Z} E_t \epsilon_{t+j-k}^Z = \sum_{Z \in \mathcal{Z}} \sum_{k=0}^{j} M_k^{X,Z} \epsilon_{t+j-k}$$

The variance of the j-steps ahead model-implied forecast error then follows, under the assumption of i.i.d innovations over time and across follows as a function of the step-size j, the Jacobian $G^{X,Z}$, the MA representation of the aggregate shock dZ, and the variance of the innovations processes:

$$\sigma_{v,X}^2(j) = \sum_{Z \in \mathcal{Z}} \sum_{k=0}^{j} M_k^2 \sigma_Z^2 = \sum_{Z \in \mathcal{Z}} \sum_{k=0}^{j} \left(\sum_s G_{k,s}^{X,Z} M_s^Z \right)^2 \sigma_Z^2.$$

[22] For analyses of the model-implied forecast error we undertake conditional on the estimated aggregate shocks parameters, particularly for backing out the counterfactual forecast errors against empirical data associated to using the behavioural vs sticky model-implied forecasts in each period covered by the sample, relying on (2) is quicker. This is because it only requires re-solving the model one more time after calibrating it at the parameter estimates without any need to simulate the paths in $d\mathbf{Z}$. Conversely, for business cycle estimation of the aggregate shocks parameters only, (1) provides a substantially quicker route as it does not require solving for G at each draw, which can instead be pre-computed while only the M^Z matrix or simulated IRF of each aggregate shock in its time-zero innovation is updated.

The cumulative (theoretical) variance of the model-implied forecast error for input X, defined by the function $\tilde{V}_X = V_X(\tilde{G}, J, \hat{\Sigma}_z, T)$, which we use for the analysis of the forecast properties comprehensively along the model horizon, is accordingly defined as:

$$\tilde{V} = \sum_{j}^{T} \sigma^2_{v,x}(j)$$

Clearly the above variance is weakly decreasing in (i) the step-size for the prediction, and (ii) the number of shocks with non-zero variance. The variance of the model-implied forecast error, conditional on the behavioural versus nested sticky expectations process encoded in the SSJ Jacobian via the proposed solvent, can thus be reduced if the estimated variances of the shocks σ^2_Z or their persistence M^Z_s are reduced. We note however that the in practice these also depend on the expectations process employed in the model estimation, so that the variance of the j-step ahead error is a highly non-linear function of the expectational parameters. For this reason, we rely on simulations to quantify, based on the above, the econometric properties of the model implied forecast.

Empirical or *counterfactual* model-implied forecast error

The model-implied forecast error variance defined above is *theoretical* in the sense that it uses the (model-conditional) estimate of the variances of the unobservable shock processes ϵ^Z for $z \in \mathcal{Z}$, i.e. information about the estimated distribution from which unobserved Gaussian innovations in the dataset are drawn. As such, it provides an assessment of the econometric properties of the model-implied forecast under alternative expectations processes, nested in our behavioural expectations process and SSJ solvent, which is not contingent on the historical path of the economy. We think a separate but related evaluation criterion to shed light on the econometric properties of the model-implied forecast comes from counterfactual analysis. Specifically, how *would* the alternative expectations models *have performed* if they had been used for macroeconomic along the historical US business cycle? In other words, what is the distribution and particularly the variance of the forecast errors the models *would have generated* against actual data if used for macroeconomic projections in the past?

To answer this, we rely on the same logic in the above section where the expression for the $j-$steps ahead projection now describes the econometrician or forecaster's "optimal" prediction policy for output at time t *in the sample* on period $t+j$, for $j < T-t$ such that we can measure the sample-based forecast-error associated to each such projection. The expression for the forecast error clearly remains the same. However, quantifying ex-post the forecast error (as opposed to moments of its theoretical distribution as above) requires knowledge of the history of innovations, which we assume the econometrician to know perfectly through to the time at which the forecast is produced. To retrieve these, we rely on the historical business cycle decomposition – conditional on the model and behavioural solvent parameter, as well as business cycle shocks estimates – to estimate the most likely path of historical innovations underlying the US business cycle series over the sample period. In practice, once the above MA representation is available, this involves solving an OLS problem, as detailed in Auclert et al. (2020), Auclert et al. (2021), yielding sequences of the innovations over the sample period.[23] Letting these estimated historical shock sequences $\hat{\epsilon}_Z$ – which are plotted in the appendix together with the business cycle decomposition for both the behavioural and sticky models – the *empirical* or *counterfactual* model-implied forecast error is accordingly defined as:

$$\hat{v}_X(t, j, \mathbf{Y}, \tilde{G}^{X,Z}_{k,z}) = \sum_{Z \in \mathcal{Z}} \sum_{k=0}^{j} \sum_{s} G^{X,Z}_{k,s} M^Z_s \hat{\epsilon}^Z_{t+j-k}.$$

The counterfactual model-implied forecast-errors computed above, for all t and $j < T-t$, are functions of the behavioural expectations and the restricted sticky models via the SSJ solvent and of the business cycle $\mathbf{Y}$. As a computational sanity and consistency check, these must be (trivially) zero for current and past-dated inputs at each time t. This is exemplified by the forecast-error plots in the

[23] This follows either from the projection theorem or from analytical maximisation of the data likelihood under linearity of the MA representation in the shock sequence. Both of them give rise to the normal equations familiar from OLS.

appendix. To quantify the econometric properties of the model-implied forecasts, we are interested in the distributions of $\hat{v}_X$ and moments thereof. As part of the exercises, we define an estimator for the (pooled) root-mean squared error of the model-forecast against data, where n_y is the number of observation periods in the sample:

$$r(\mathbf{Y}, \tilde{G}_{k,z}^{X,Z}) = \frac{1}{d} \sum_{t \leq n_y} \sum_{j \leq n_y} v_X(t,j)^2$$

$$d = n_y^2 - 2^{-1} n_y (n_y + 1).$$

The denominator d applies a bias-correction to account for the number of non-trivial forecast errors, as by definition the forecast error is only meaningfully defined for forecasts of *strictly* future inputs at each horizon t in the dataset, and trivially zero for forecast of current or future variables. This is done with Gauss' formula noting that if we stack $v_X(t,j)$ in a $T \times T$ matrix with the (t,j) entry equal to the (ax-post) error in the time-t forecast on the period-j input, we are interested in the upper triangular section of the matrix excluding the main diagonal, which is given by d under the observation as the matrix is square.